INCH-POUND

MIL-DTL-32771
11 June 2024

DETAIL SPECIFICATION

FITTINGS, BRAZE-ON AND REUSABLE TYPE – FOR FLEXIBLE CONDUIT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes GR2000-series (braze-on type) transition fittings and accessory fittings and RP2000-series (reusable type) fittings to be used on the flexible metal shielding conduit specified in MIL-PRF-32770, in which the dimension of the outside diameter (OD) for each nominal size of conduit is standardized. This specification is intended to be a companion document for NAVSEA S9407-AB-HBK-010, Handbook of Shipboard Electromagnetic Shielding Practices.

1.2 Classification. Fittings covered by this specification are of the following series, as specified (see 6.2). The use of GR2000-series or RP2000-series fittings is optional. However, for most installations, the RP2000-series fittings are preferred because they require no brazing for installation on the conduit, and a greater selection of adapters is available.

a. GR2000-series – GR2000-series fittings (braze-on type) require that a component of the fitting be brazed directly to the conduit braid. The assembly is made watertight by means of O-ring seals.

b. RP2000-series – RP2000-series fittings (reusable type) are attached to the conduit by gripping the conduit braid directly by means of friction and do not provide a watertight assembly.

1.3 Part or identifying number (PIN). The basic PIN is constructed as follows followed by a unique set of characteristics for each fitting as specified on figures 1 through 35:

Basic PIN:	**M**	**32771**	-	**XXXXXX**
	Prefix for Military Specification	Specification Number	-	Series (see the Basic Part No. of tables I and II)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

L-P-410 - Plastic, Polyamide (Nylon), Rigid: Rods, Tubes, Flats, Molded and Cast Parts

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-1310 - Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9407-AB-HBK-010 - Handbook of Shipboard Electromagnetic Shielding Practices

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM B85/B85M - Standard Specification for Aluminum Alloy Die Castings

ASTM D4066 - Standard Classification System for Nylon Injection and Extrusion Materials (PA)

ASTM E9 - Standard Test Methods of Compression Testing of Metallic Materials at Room Temperature

(Copies of these documents are available online at www.astm.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS A5.8 - Specification for Filler Metals for Brazing and Braze Welding

(Copies of this document are available online at www.aws.org.)

SAE INTERNATIONAL

SAE AIR4127 - Steel: Chemical Composition and Hardenability

SAE AMS2404 - Plating, Electroless Nickel

SAE AMS2700 - Passivation of Corrosion Resistant Steels

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SAE AMS-QQ-A-225	-	Aluminum and Aluminum Alloy, Bar, Rod, Wire, or Special Shapes; Rolled, Drawn, or Cold Finished; General Specification for
SAE AMS-QQ-P-416	-	Plating, Cadmium (Electrodeposited)
SAE AMS-QQ-S-763	-	Steel, Corrosion Resistant, Bars, Wires, Shapes, and Forgings
SAE AS28775	-	Packing, Performed – MS28775 O-Ring
SAE AS39029	-	Connectors, Electrical, Crimp Removable, Series MS3400
SAE AS50151	-	Connectors, Electrical, Circular Threaded, AN Type, General Specification for

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Construction.

3.1.1 GR2000-series.

3.1.1.1 Function. A transition fitting shall provide the physical connection between flexible metal conduit and an electrical connector or fitting. An accessory fitting shall provide the physical connection between flexible conduit and a bulkhead or panel and provide the means for coupling two lengths of conduit together.

When the GR2000-series type of transition fitting is used, stainless steel cylindrical end fittings shall be brazed directly to the braid of the flexible conduit. The other components of the transition fitting shall be attached to the end fitting with a ferrule that operates by friction. The assembly shall be made watertight by O-ring seals.

By comparison, the RP2000-series reusable fittings utilize a ferrule that grips the conduit braid directly by friction and does not provide for a watertight assembly.

3.1.1.2 Dimensions. All dimensions and tolerances shall be as specified on the figures listed in table I.

TABLE I. GR2000-series transition and accessory fittings.

Figure No.	Basic Part No.	Description
1	GR2120	Transition fitting for SAE AS50151 connectors
2	GR2121	Enclosure fitting
3	GR2122	90° elbow, non-environmental (style A)
4	GR2122	90° elbow, environmental (style B)
5	GR2123	Transition fitting for bulkhead stuffing tube
6	GR2124	Transition fitting for MS series backshell
7	GR2125	Transition fitting for triaxial connector
8	GR2126	End fitting for flexible conduit
9	GR2127	Conduit coupling
10	GR2128	Panel fitting
11	GR2129	Ferrule and coupling nut for flexible conduit
12	GR2130	Isolator adapter

3.1.1.3 Minimum wall thickness. The minimum wall thickness for the body of the GR2120, GR2123, GR2124, and GR2125 fittings shall be 1/16 inch.

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3.1.2 RP2000-series.

3.1.2.1 Function. The RP2000-series reusable fittings shall provide a method of joining connectors, applicable hull fittings, and similar devices to flexible shielding conduit. There are three basic types in this series:

a. An end fitting, which is attached to a conduit and provides a standard thread size for attaching adapters.

b. A coupling, which typically joins two lengths of conduit.

c. An adapter, which provides thread compatibility between an end fitting and various devices such as connectors and hull fittings.

The term “reusable” means that an end fitting or coupling can be readily attached to or removed from the conduit simply by tightening or loosening a gripping mechanism.

Table II lists the RP2000-series fittings. These fittings shall be as specified on figures 13 through 35.

TABLE II. RP2000-series fittings.

Type	Figure No.	Basic Part No.	Description
End fittings	13	RP2100	Conduit end fitting, straight
	14	RP2110	Conduit end fitting, 90°
	15	RP2120	Conduit end fitting, 45°
	16	RP2130	End fitting, conduit-to-panel
	17	RP2140	End fitting, conduit-to-pipe thread (tapered)
	18	RP2141	End fitting, conduit-to-pipe thread (straight)
Couplings	19	RP2200	Coupling, conduit-to-conduit
	20	RP2210	Coupling, conduit-to-pipe
Adapters (connectors)	21	RP2311	Adapter for SAE AS50151 connector
	22	RP2321	Adapter for triaxial connector
	23	RP2330	Adapter for coaxial connector
	24	RP2340	Adapter for MIL-DTL-26482 connector
	25	RP2350	Adapter for SAE AS85049 type connectors
	26	RP2360	Adapter for MIL-DTL-28840 backshell
Adapters (miscellaneous)	27	RP2400	Adapter for MIL-S-24235/1 stuffing tube
	28	RP2411	Adapter for miscellaneous threaded fittings
	29	RP2420	Adapter for MIL-DTL-24231 hull fitting
	30	RP2430	Adapter for pipe thread (tapered)
	31	RP2431	Adapter for pipe thread (straight)
	32	RP2440	Adapter, nonmetallic
Adapter (end)	33	RP2500	Adapter for panel termination
Adapters (angle)	34	RP2610	Adapter, 90°
	35	RP2620	Adapter, 45°

3.1.2.2 Basic construction. Figures 13 through 35 show the basic construction and the essential characteristics of the RP2000-series fittings. The figures are those of typical fittings and are intended to show function only. Other constructional details are left to the option of the manufacturer except as noted elsewhere in this specification.

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Welded fabrication of the fittings is allowed. Dimensions and thread sizes shown are those that shall exist after plating is completed.

3.1.2.3 Conduit gripping mechanism. The end fittings and couplings on figures 13 through 20 show a slotted ferrule as the conduit-gripping mechanism, but any other simple method that meets the performance requirements is acceptable.

3.1.2.4 Overlap length. The overlap length "L" in the figures of end fittings and couplings shall be not less than the minimum length of conduit required by the conduit-gripping mechanism to function properly.

3.1.2.5 Minimum inside diameter. The minimum inside diameter (ID) of end fittings and of conduit couplings shall be not less than the minimum ID of the associated conduit.

3.1.2.6 Knurl length. The total length of a knurl shall consist of either one wide band (as shown on the figures) or two or more narrow bands of knurls located close to each other.

3.1.2.7 Standard thread size. Table III lists a series of standard thread sizes that have been selected for mating between end fittings and adapters. A single thread size is associated with each size of end fitting, and each size of end fitting is associated with only one size of conduit. Thus, any size conduit can be terminated at a variety of connectors, hull fittings, etc., by means of its associated end fitting and the proper adapter. The threads shall be class 2A and 2B for the male and female threads, respectively. They shall be constructed according to the standards set forth in FED-STD-H28.

TABLE III. Standard thread sizes for RP2000-series end fittings and adapters.

Standard Size Conduit			End Fitting			Adapter		
Conduit Part Number	Conduit Size (Nominal ID) (inches)	Conduit OD (Over Braid)	B Thread (Class 2B)			C Thread (Class 2A)		
EM02	¼	0.42±0.01	0.4375	-28	UNEF	0.4375	-28	UNEF
EM03	⅜	0.54±0.01	0.5625	-24	UNEF	0.5625	-24	UNEF
EM04	½	0.69±0.01	0.6875	-24	UNEF	0.6875	-24	UNEF
EM05	⅝	0.82±0.01	0.8125	-20	UNEF	0.8125	-20	UNEF
EM06	¾	0.94±0.01	0.9375	-20	UNEF	0.9375	-20	UNEF
EM08	1	1.25±0.01	1.250	-18	UNEF	1.250	-18	UNEF
EM10	1¼	1.50±0.01	1.5625	-18	UNEF	1.5625	-18	UNEF
EM12	1½	1.75±0.01	1.875	-16	UN	1.875	-16	UNEF
EM16	2	2.28±0.015	2.375	-16	UN	2.375	-16	UNEF
EM20	2½	2.78±0.015	2.875	-16	UN	2.875	-16	UN
EM24	3	3.28±0.015	3.375	-16	UN	3.375	-16	UN

3.2 Materials.

3.2.1 GR2000-series.

3.2.1.1 Components material. All components of the fittings, except for the GR2126 end fitting and the GR2129 ferrule, shall be made of mild steel, AISI-type B1113, or any AISI-type low-carbon, hot- or cold-rolled steels, in accordance with SAE AIR4127, having chemical compositions within the following ranges:

a. Carbon (C): 0.08 to 0.25 percent max.

b. Manganese (Mn): 0.25 to 1.15 percent max.

c. Phosphorus (P): 0.04 to 0.12 percent max.

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d. Sulphur (S): 0.05 to 0.35 percent max.

e. Lead (Pb): 0.15 to 0.35 percent max.

3.2.1.2 End fitting (GR2126) and ferrule (GR2129) material. The GR2126 end fitting and GR2129 ferrule shall be made of corrosion-resistant steel (CRES), 300 series, in accordance with SAE AMS-QQ-S-763.

3.2.1.3 Conduit coupling (GR2127) material. The GR2127 conduit coupling shall be made of low-carbon steel (see 3.2.1.1).

3.2.1.4 Joining material. Any joining material shall have the basic alloy content of the parts being joined. Brazing material shall be in accordance with AWS A5.8 for FS class joints.

3.2.1.5 O-ring gasket material. O-ring gaskets that are part of fittings shall be in accordance with SAE AS28775, class B.

3.2.1.6 Isolator adapter (GR2130) material. The GR2130 isolator adapter shall be made of nylon (see table 12-2, of figure 12).

3.2.2 RP2000-series.

3.2.2.1 Material for most RP2000-series fittings. Unless otherwise specified herein, the RP2000-series fittings shall be made of mild steel, AISI-type B113, or any AISI-type low-carbon, hot- or cold-rolled steels, in accordance with SAE AIR4127, having chemical compositions within the following ranges:

a. Carbon (C): 0.08 to 0.25 percent max.

b. Manganese (Mn): 0.25 to 1.15 percent max.

c. Phosphorus (P): 0.04 to 0.12 percent max.

d. Sulfur (S): 0.05 to 0.35 percent max.

e. Lead (Pb): 0.15 to 0.35 percent max.

3.2.2.2 Optional material for RP2330 adapter. The RP2330 adapter shall be fabricated from polyamide (nylon) in accordance with L-P-410 or ASTM D4066 whenever installations require that the RP2330 insulate the connector from conduit.

3.2.2.3 Optional material for RP2350 adapter. The RP2350 adapter shall be fabricated from aluminum alloy in accordance with SAE AMS-QQ-A-225 or ASTM B85/B85M, provided that minimum spacing requirements for aluminum backshells as specified in S9407-AB-HBK-010 are met.

3.2.2.4 Material for RP2440 adapter. The RP2440 adapter shall be fabricated from polyamide (nylon) rods or tubes in accordance with L-P-410 or molded from nylon molding plastic in accordance with ASTM D4066. An optional material is the “super tough” nylon identified by the Dupont trade name “Zytel” ST801 (or equal). Super tough nylon comes in stock shapes for machining, or it can be molded.

3.3 Finish.

3.3.1 GR2000-series.

3.3.1.1 Finish for GR2000-series low-carbon steel fittings. Finish for components made of low-carbon steel shall be a nickel coating in accordance with SAE AMS2404.

3.3.1.2 Finish for GR2000-series CRES fittings. Parts made of 300-series CRES shall be passivated in accordance with SAE AMS2700.

3.3.1.3 Finish for conduit coupling (GR2127). Finish for GR2127 conduit coupling shall be a cadmium plating in accordance with SAE AMS-QQ-P-416, type I, class 3.

3.3.2 RP2000-series.

3.3.2.1 Finish for steel fittings. The finish on the steel fittings shall be electroless nickel in accordance with SAE AMS2404.

3.3.2.2 Finish for aluminum fittings. The finish on the aluminum fitting shall be electroless nickel in accordance with SAE AMS2404.

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3.4 Workmanship. Fittings shall be fabricated so that they are uniform in quality and free from defects that could make the fitting unsatisfactory for the intended purpose.

3.4.1 Welding or brazing. Fittings fabricated by welding or brazing shall have joints free from defects such as cracks, undercuts, and gaps. There shall be no burn-through. Fillets shall be uniform and smooth.

3.4.2 Cable protection. The fittings shall have smooth inner surfaces free from sharp edges that could abrade the insulation or braid of a cable. Use chamfers or radii where applicable. Break all sharp edges and corners.

3.5 Marking. Each GR2000-series and RP2000-series fitting shall be marked in accordance with MIL-STD-130, with the name of the manufacturer and with its complete PIN as specified herein. Additional markings may be included at the option of the manufacturer. Additionally, a printed label or tag shall be furnished with each RP2000-series fitting. The label shall provide the PIN, the manufacturer's name, and any other information, such as a sketch or instruction, that the manufacturer deems pertinent for proper installation.

3.6 Performance.

3.6.1 Mechanical. The body of a transition fitting shall telescope over the conduit end fitting (GR2126) to allow room for cable termination. Assembly of parts shall not place any torque or tensile loads on the transition fitting.

3.6.2 Moisture sealing. When moisture sealing or splashproof requirements are specified (see 6.2), an O-ring gasket of material specified in 3.2.1.5 shall be provided as part of the transition fitting to seal each critical joint.

Where moisture sealing is not a requirement, O-ring gaskets may be omitted from transition fittings. However, such omission shall not impair the mechanical performance of the assembly.

3.6.3 Cable protection. The GR2000-series fitting shall have smooth inner surfaces free of sharp burrs or edges that could abrade the insulation or shield of a cable.

3.6.4 Welding restriction. Installation of any RP2000-series fitting shall not require welding, brazing, or similar operations.

3.6.5 Tensile and compressive load. GR2000-series transition fittings and end fittings and RP2000-series end fittings and couplings shall withstand, without slippage, a tensile load of 50 pounds applied between fitting and conduit for 1 minute in accordance with ASTM A370. Also, such fittings shall withstand, without slippage, a compressive load of 50 pounds applied between fitting and conduit for 1 minute in accordance with ASTM E9.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the examination of 4.2 and the test of 4.3.

4.2 Examination. Fittings shall be visually and dimensionally examined to determine conformance with the construction (see 3.1), finish (see 3.3), workmanship (see 3.4), marking (see 3.5), and performance (see 3.6.1 through 3.6.4) requirements herein. Material (see 3.2) requirements shall be verified as specified (see 6.2).

4.3 Tensile and compressive load test. Fittings and couplings shall be tested in accordance with ASTM A370 and ASTM E9 to ensure compliance with tensile and compressive load requirements of 3.6.5.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document is intended as a support document for S9407-AB-HBK-010. As such this document indirectly supports the installation of electrical cabling so that electromagnetic compatibility (EMC) of these systems is ensured. Due to the unique nature of the submarine environment, the primary focus herein is for U.S. Navy submarines; however, the guidance provided can be utilized in many other naval installations.

This document is applicable to new construction, as well as all currently active platforms. As such, this document maintains "forward and backward" applicability. This is needed due to the fact that active platforms still undergo modifications and changes where the guidance provided herein is applicable to their respective cabling installations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Series required (see 1.2).

c. When moisture sealing or splashproof requirements are needed (see 3.6.2).

d. Requirement for material verification (see 4.2).

e. Packaging requirements (see 5.1).

6.3 PIN cross-reference. Existing stock of fittings with PINs in accordance with S9407-AB-HBK-010, without the M32771 prefix, are substitutable for fittings with the M32771 prefix in accordance with MIL-DTL-32771.

6.4 Specification cross-reference. Table IV is provided for backfit installation purposes and cross-references legacy specifications with their respective superseding document.

TABLE IV. Specification cross-reference.

Legacy Specification	Updated Specification	Table/Figure
MS3155, Connector, Electric, Rear Accessory Design Standard	SAE AS85049, Connector Accessories, Electrical, General Specification for	Table II, figure 25
MIL-T-20157, Tube and Pipe, Carbon Steel, Seamless	MIL-P-24691, Tube and Pipe, Carbon Alloy and Stainless Steel, Seamless and Welded, General Specification for	Figure 20
MIL-C-5015, Connectors, Electrical, Circular Threaded, AN Type, General Specification for	SAE AS50151, Connectors, Electrical, Circular Threaded, AN Type, General Specification for	Table II, figure 21
MIL-C-81703, Connectors, Electric Circular, Miniature, Rack and Panel or Push-Pull Coupling, Environment Resisting	SAE AS81703, Connectors, Electric, Circular, Miniature, Rack and Panel or Push-Pull Coupling, Environment Resisting	Figure 25
NAS1599, Connectors, General Purpose, Electrical, Miniature Circular, Environment Resisting, 200 Degrees C Maximum Temperature	MIL-DTL-83723, Connectors, Electrical, Circular, Environment Resisting and Hermetically Sealed, Receptacles and Plugs, General Specification for	Figure 25
MIL-M-20693, Molding Plastic, Polyamide (Nylon), Rigid	ASTM D4066, Standard Classification System for Nylon Injection and Extrusion Materials	Figure 32

6.5 Subject term (key word) listing.

Connectors

Electromagnetic Compatibility

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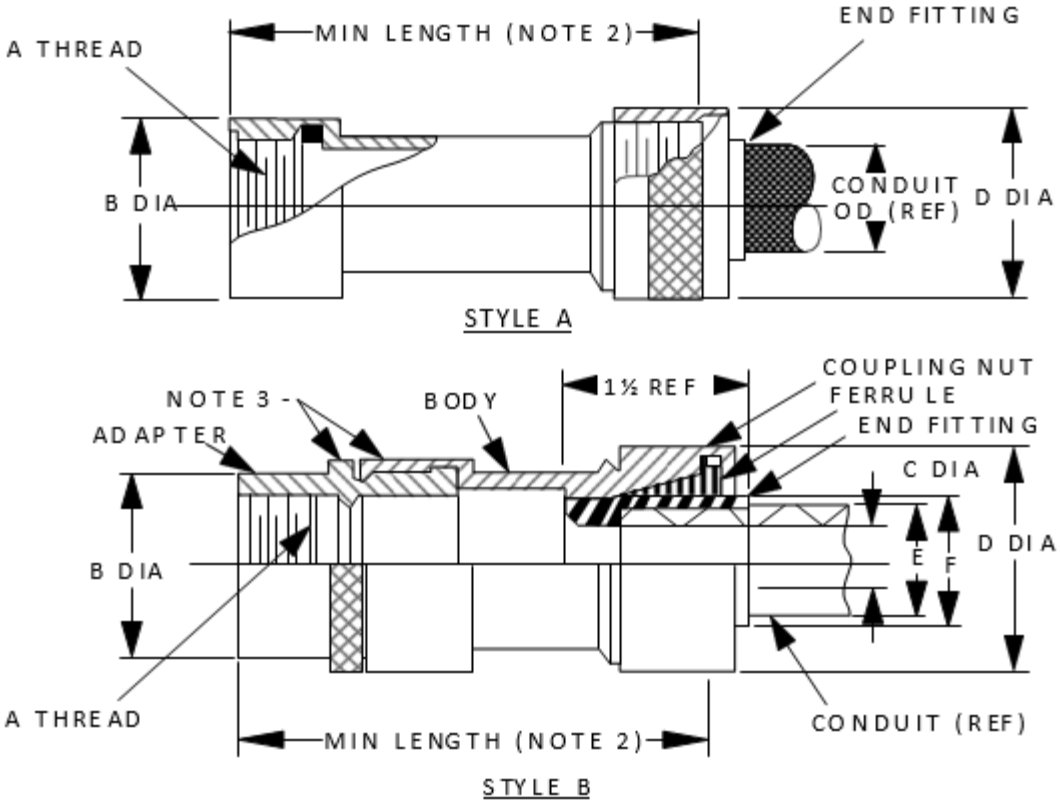


FIGURE 1. GR2120 transition fitting for SAE AS50151 connector.

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TABLE 1-1. Connector and adapter combinations for GR2120.

Connector		Adapter End	
Conn. Shell Size	**Conn. Symbol (See table 1-3)**	**A Thread Class 2B**	**B Dia. +0.010 -0.020**
8S	L	⅜ - 32 UNEF	0.562
8S	H	7/16 - 27 UNS	0.625
8	JMN	7/16 - 28 UNEF	0.625
10S	HJLN	½ - 28 UNEF	0.687
10S	M	9/16 - 24 UNEF	0.687
10SL	MN	9/16 - 24 UNEF	0.687
10SL	HJI	7/16 - 28 UNEF	0.812
12S	LN	⅝ - 24 UNEF	0.812
12S	HJM	⅝ - 24 UNEF	0.812
14S	HJLMN	11/16 - 24 UNEF	0.937
16S	HJLMN	¾ - 20 UNEF	1.062
12	LN	⅞ - 20 UNEF	0.812
12	HM	⅝ - 24 UNEF	0.812
14	HLMN	11/16 - 24 UNS	0.937
16	HJLMN	⅞ - 20 UNEF	1.062
18	HJMN	1 - 20 UNEF	1.187
20	J	1⅛ - 24 UNS	1.312
20	HLMN	1⅛ - 18 UNEF	1.312
22	HJLMN	1¼ - 18 UNEF	1.437
24	HJLMN	1⅜ - 18 UNEF	1.562
28	HJLMN	1⅝ - 18 UNEF	1.812
32	LMN	1⅞ - 16 UN	2.062
32	HJ	1 29/32 - 18 UNS	2.062
36	L	2 1/16 - 16 UN	2.312
36	J	2 1/16 - 20 UNEF	2.312
36	MN	2⅛ - 16 UN	2.312
36	H	2⅛ - 18 UNS	2.312
40	L	2 5/16 - 16 UN	2.562
40	HMN	2⅜ - 16 UN	2.562
44	HN	2⅝ - 16 UN	2.875
48	N	2 3/16 - 18 UNS	3.125
48	H	2⅞ - 16 UN	3.125

FIGURE 1. GR2120 transition fitting for SAE AS50151 connector – Continued.

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TABLE 1-2. GR2120 sizes.

Standard Size Conduit			Transition Fitting				
Conduit Part Number	Conduit Size (Nom. ID)	OD (Over Braid)	End Fitting			Nut	Dash No.
			C Dia.	E +0.010 -0.000	F ±0.005	D Max.	
EM02	¼	0.42±0.01	0.25	0.450	0.530	1.03	EM02
EM03	⅜	0.54±0.01	0.38	0.570	0.650	1.23	EM03
EM04	½	0.69±0.01	0.50	0.720	0.800	1.36	EM04
EM05	⅝	0.82±0.01	0.62	0.850	0.930	1.59	EM05
EM06	¾	0.94±0.01	0.75	0.970	1.050	1.73	EM06
EM08	1	1.25±0.01	1.00	1.280	1.360	1.91	EM08
EM10	1¼	1.50±0.01	1.25	1.530	1.610	2.28	EM10
EM12	1½	1.75±0.01	1.50	1.780	1.860	2.66	EM12
EM16	2	2.28±0.015	2.00	2.315	2.395	3.17	EM16
EM20	2½	2.78±0.015	2.50	2.815	2.895	3.97	EM20
EM24	3	3.28±0.015	3.00	3.315	3.395	4.35	EM24

TABLE 1-3. GR2120 symbols.

SAE AS50151 Connector			
Symbol	Mfr.	Class	Description
H	AMPHENOL	A	MS3100, MS3101, MS3106, 97-3100, 97-3101, 97-3106
J	AMPHENOL	E, R	MS3100, MS3101, MS3106, 69-3100, 69-3101, 69-3106
L	BENDIX	A, E, R	MS3100, MS3101, MS3106, 10-214, 10-720, 10-726
M	CANNON	A	MS3100, MS3101, MS3106
N	CANNON	E, R	MS3100, MS3101, MS3106, (CAO1, CAO6)

FIGURE 1. GR2120 transition fitting for SAE AS50151 connector – Continued.

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GR2120 PIN Construction:

Example:	**M32771-GR2120**	-	**M**	**12**	**EM08**	**2.5**
	Basic PIN	-	Connector Symbol (see table 1-3)	Connector Shell Size (see table 1-1)	Dash Number (see table 1-2)	Body/Adapter Length (see note 2)

NOTES:

1. Material shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404, except end fitting shall be CRES, 300 series, passivated.
2. Standard minimum length shall be 2 inches for one-piece (style A) and 2.5 inches for two-piece (style B).
3. When conduit OD exceeds ID of connector shell, a two-piece fitting (style B) shall be supplied.
4. Assembly identified in accordance MIL-STD-130.
5. Discard connector rear hardware.
6. Basic PIN without O-ring is M32771-GR2120; basic PIN with O-ring is M32771-GRO2120.
7. Dimensions are in inches.
8. Manufacturer information provided for historical reference only. Information for procuring material should be obtained directly from vendor.

FIGURE 1. GR21120 transition fitting for SAE AS50151 connector – Continued.

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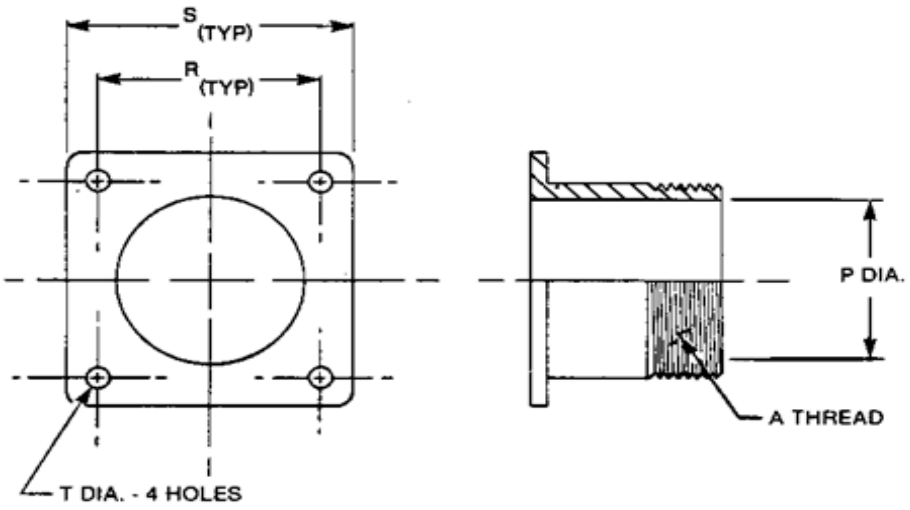


TABLE 2-1. GR2121 sizes.

PIN	Shell Size	A Thread Class 2A	P Dia. Max.	R ±0.005	S Max.	T Dia.
M32771-GR2121-8S	8S	½ - 28 UNEF	0.364	0.594	0.885	0.120
M32771-GR2121-10S	10S	½ - 28 UNEF	0.364	0.718	1.005	0.120
M32771-GR2121-10SL	10SL	⅝ - 24 NEF	0.443	0.718	1.005	0.120
M32771-GR2121-12S	12S	⅝ - 24 NEF	0.443	0.813	1.104	0.120
M32771-GR2121-14S	14S	¾ - 20 UNEF	0.567	0.906	1.198	0.120
M32771-GR2121-16S	16S	⅞ - 20 UNEF	0.692	0.968	1.291	0.120
M32771-GR2121-18	18	1 - 20 UNEF	0.804	1.062	1.406	0.120
M32771-GR2121-20	20	1 3/16 - 18 NEF	0.916	1.156	1.531	0.120
M32771-GR2121-22	22	1 3/16 - 18 NEF	0.924	1.250	1.656	0.120
M32771-GR2121-24	24	1 7/16 - 18 NEF	1.173	1.375	1.781	0.147
M32771-GR2121-28	28	1 7/16 - 18 NEF	1.182	1.562	2.031	0.147
M32771-GR2121-32	32	1¾ - 18 NS	1.485	1.750	2.281	0.173
M32771-GR2121-36	36	2 - 18 NS	1.734	1.937	2.531	0.173
M32771-GR2121-40	40	2¼ - 16 UN	1.976	2.187	2.781	0.173
M32771-GR2121-44	44	2½ - 16 UN	2.235	2.375	3.031	0.173
M32771-GR2121-48	48	3 - 16 UN	2.729	2.625	3.281	0.173

NOTES:
1. Material shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404.
2. Assembly identified in accordance MIL-STD-130.
3. Dimensions are in inches.
4. According to FED-STD-H28, the American National thread series (N) has been superseded by the United thread series (UN). These include the NEF and NS thread sizes, which are retained for use with connectors having NEF and NS threads.

FIGURE 2. GR2121 enclosure fitting.

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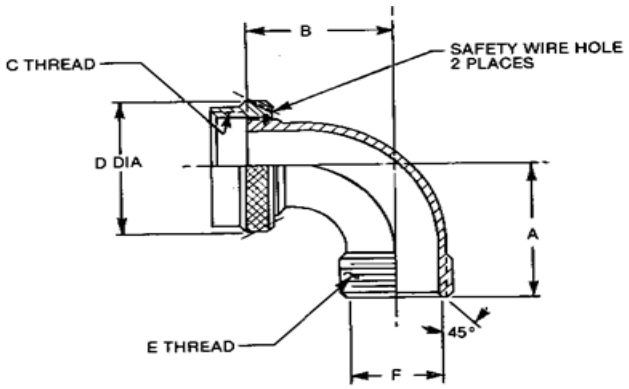


TABLE 3-1. GR2122 style A sizes.

PIN	Shell Size	A ±0.094	B ±0.094	C Thread Class 2B	D Dia. ±0.010	E Thread Class 2A	F Dia.
M32771-GR2122-101	10S	0.735	0.765	½ - 28 UNEF	0.722	½ - 28 UNEF	0.364
M32771-GR2122-121	12S	0.892	0.844	⅝ - 24 UNEF	0.866	⅝ - 24 UNEF	0.490
M32771-GR2122-131	12	0.892	0.844	⅝ - 24 UNEF	0.866	⅝ - 24 UNEF	0.490
M32771-GR2122-141	14S	1.016	0.968	¾ - 20 UNEF	1.010	¾ - 20 UNEF	0.604
M32771-GR2122-151	14	1.016	0.968	¾ - 20 UNEF	1.010	¾ - 20 UNEF	0.604
M32771-GR2122-161	16S	1.173	1.125	⅞ - 20 UNEF	1.155	⅞ - 20 UNEF	0.729
M32771-GR2122-171	16	1.173	1.125	⅞ - 20 UNEF	1.155	⅞ - 20 UNEF	0.729
M32771-GR2122-181	18	1.391	1.312	1 - 20 UNEF	1.362	1 - 20 UNEF	0.854
M32771-GR2122-201	20	1.485	1.406	1⅛ - 18 UNEF	1.487	1⅛ - 18 UNEF	0.972
M32771-GR2122-202	20	1.485	1.406	1⅛ - 18 UNEF	1.487	1 3/16 - 18 UNEF	1.034
M32771-GR2122-221	22	1.609	1.530	1¼ - 18 UNEF	1.612	1¼ - 18 UNEF	1.097
M32771-GR2122-222	22	1.609	1.530	1¼ - 18 UNEF	1.612	1⅞ - 18 UNEF	1.034
M32771-GR2122-241	24	1.798	1.719	1⅜ - 18 UNEF	1.737	1⅜ - 18 UNEF	1.222
M32771-GR2122-242	24	1.798	1.719	1⅜ - 18 UNEF	1.737	1 7/16 - 18 UNEF	1.284
M32771-GR2122-281	28	2.079	2.016	1⅝ - 18 UNEF	1.987	1⅝ - 18 UNEF	1.457
M32771-GR2122-282	28	2.263	2.016	1⅝ - 18 UNEF	1.987	1 7/16 - 18 UNEF	1.284
M32771-GR2122-321	32	2.266	2.234	1⅞ - 16 UN	2.237	1⅞ - 16 UN	1.738
M32771-GR2122-322	32	2.266	2.234	1⅞ - 16 UN	2.237	1¾ - 18 UNEF	1.566
M32771-GR2122-361	36	2.610	2.609	2 1/16 - 16 UN	2.425	2 1/16 - 16 UN	1.925
M32771-GR2122-362	36	2.610	2.609	2 1/16 - 16 UN	2.425	2 - 18 UNS	1.816

FIGURE 3. GR2122 90° elbow, non-environmental (style A).

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TABLE 3-1. GR2122 style A sizes - Continued.

PIN	Shell Size	A ±0.094	B ±0.094	C Thread Class 2B	D Dia. ±0.010	E Thread Class 2A	F Dia.
M32771-GR2122-401	40	2.625	2.625	2 5/16 - 16 UNS	2.674	2½ - 16 UN	2.174
M32771-GR2122-402	40	2.625	2.625	2⅜ - 16 UN	2.737	2½ - 16 UN	2.174
M32771-GR2122-441	44	2.875	2.875	2⅝ - 16 UN	2.987	2⅝ - 16 UN	2.424
M32771-GR2122-481	48	3.125	3.125	2 13/16 - 18 UNS	3.174	3 - 16 UN	2.674
M32771-GR2122-482	48	3.125	3.125	2⅞ - 16 UN	3.237	3 - 16 UN	2.674
M32771-GR2122-101	10S	0.735	0.765	½ - 28 UNEF	0.722	½ - 28 UNEF	0.364

NOTES:
1. Material shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404.
2. Assembly identified in accordance with MIL-STD-130.
3. Dimensions are in inches.

FIGURE 3. GR2122 90° elbow, non-environmental (style A) - Continued.

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TABLE 4-1. GR2122 style B symbols.

SAE AS50151 Connector			
Symbol	**Mfr.**	**Class**	**Series**
H	AMPHENOL	A	MS3100, MS3106, 97-3100, 97-3106
J	AMPHENOL	R	MS3100, MS3106, 97-3100, 97-3106
L	BENDIX	A	MS3100, MS3106
L	BENDIX	E, R	MS3100, MS3106, 10-214, 10-721, 10-726
M	CANNON	A	MS3101
N	CANNON	E, R	MS3100, MS3106, CA00, CA01, CA06

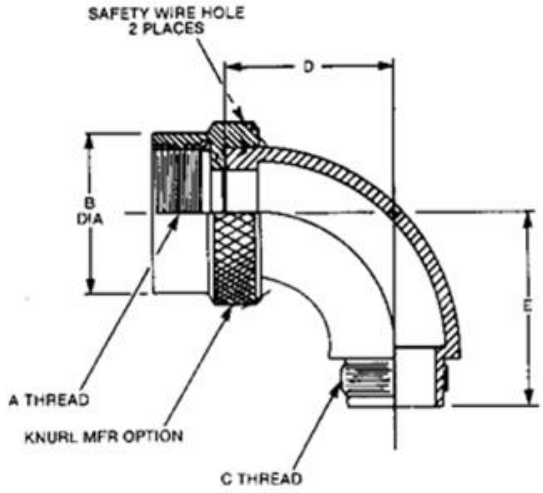


TABLE 4-2. GR2122 style B sizes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:43Z
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PIN	Shell Size	Conn.	A Thread Class 2B	B Dia.	C Thread Class 2A	D ±0.094	E ±0.094
M32771-GR2122-081E	8S	L	⅜ - 32 UNEF	0.562	½ - 28 UNEF	1.406	1.609
M32771-GR2122-082E	8S	H	7/16 - 27 UNS	0.625	½ - 28 UNEF	1.406	1.609
M32771-GR2122-083E	8S	JMN	7/16 - 28 UNEF	0.625	½ - 28 UNEF	1.406	1.609
M32771-GR2122-101E	10S	HJLN	½ - 28 UNEF	0.687	½ - 28 UNEF	1.406	1.609
M32771-GR2122-102E	10S	M	9/16 - 24 UNEF	0.687	⅝ - 24 UNEF	1.406	1.609
M32771-GR2122-103E	10SL	MN	9/16 - 24 UNEF	0.687	⅝ - 24 UNEF	1.406	1.609
M32771-GR2122-104E	10SL	HJL	⅝ - 24 UNEF	0.812	⅝ - 24 UNEF	1.406	1.609
M32771-GR2122-121E	12S	LN	⅝ - 24 UNEF	0.812	⅝ - 24 UNEF	1.406	1.609
M32771-GR2122-122E	12S	HJN	11/16 - 24 UNEF	0.812	¾ - 20 UNEF	1.406	1.609
M32771-GR2122-123E	12SL	H	¾ - 20 UNEF	0.937	¾ - 20 UNEF	1.406	1.609
M32771-GR2122-141E	14S	HJLMN	¾ - 20 UNEF	0.937	¾ - 20 UNEF	1.406	1.609
M32771-GR2122-161E	16S	HJLMN	⅞ - 20 UNEF	1.062	⅞ - 20 UNEF	1.406	1.609
M32771-GR2122-132E	12	LN	⅝ - 24 UNEF	0.812	⅝ - 20 UNEF	1.406	1.609
M32771-GR2122-133E	12	NM	11/16 - 24 UNEF	0.812	¾ - 20 UNEF	1.406	1.609
M32771-GR2122-151E	14	HLMN	¾ - 20 UNEF	0.937	¾ - 20 UNEF	1.406	1.609
M32771-GR2122-171E	16	HJLMN	⅞ - 20 UNEF	1.062	⅞ - 20 UNEF	1.406	1.609
M32771-GR2122-181E	18	HJLMN	1 - 20 UNEF	1.187	1 - 20 UNEF	1.781	1.984

FIGURE 4. GR2122 90° elbow, environmental (style B).

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TABLE 4-2. GR2122 style B sizes – Continued.

PIN	Shell Size	Conn.	A Thread Class 2B	B Dia.	C Thread Class 2A	D ±0.094	E ±0.094
M32771-GR2122-203E	20	J	1⅛ - 24 UNS	1.312	1⅛ - 18 UNEF	1.781	1.984
M32771-GR2122-201E	20	HLMN	1⅛ - 18 UNEF	1.312	1⅛ - 18 UNEF	1.781	1.984
M32771-GR2122-221E	22	HJLMN	1¼ - 18 UNEF	1.437	1¼ - 18 UNEF	2.156	2.359
M32771-GR2122-241E	24	HJLMN	1⅜ - 18 UNEF	1.562	1⅜ - 18 UNEF	2.156	2.359
M32771-GR2122-281E	28	HJLMN	1⅝ - 18 UNEF	1.812	1⅝ - 18 UNEF	2.531	2.734
M32771-GR2122-321E	32	LMN	1⅞ - 16 UN	2.062	1⅞ - 16 UN	3.281	3.484
M32771-GR2122-323E	32	HJ	1 29/32 - 18 UNS	2.062	1⅞ - 16 UN	3.281	3.484
M32771-GR2122-361E	36	L	2 1/16 - 16 UN	2.312	2 1/16 - 16 UN	3.281	3.484
M32771-GR2122-363E	36	J	2 1/16 - 20 UNEF	2.312	2 1/16 - 16 UN	3.281	3.484
M32771-GR2122-364E	36	MN	2⅛ - 16 UN	2.312	2 1/16 - 16 UN	3.281	3.484
M32771-GR2122-365E	36	H	2⅛ - 18 UNS	2.312	2 1/16 - 16 UN	3.281	3.484
M32771-GR2122-403E	40	L	2 5/16 - 16 UN	2.562	2 1/16 - 16 UN	4.031	4.234
M32771-GR2122-404E	40	HMN	2⅜ - 16 UN	2.562	2 1/16 - 16 UN	4.031	4.234
M32771-GR2122-442E	44	HNL	2⅝ - 16 UN	2.875	2 1/16 - 16 UN	4.031	4.234
M32771-GR2122-483E	48	N	2 13/16 - 18 UNS	3.125	2 1/16 - 16 UN	4.781	4.984
M32771-GR2122-484E	48	H	2⅞ - 16 UN	3.125	2 1/16 - 16 UN	4.781	4.984

NOTES:
1. Material shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404.
2. Assembly identified in accordance with MIL-STD-130.
3. Dimensions are in inches.
4. Amphenol, Bendix, and Cannon “A” class connectors and shell size “10SL” of Cannon “E, R” class connectors shall not be selected for environmental assembly.
5. Manufacturer information provided for historical reference only. Information for procuring material shall be obtained directly from vendor.

FIGURE 4. GR2122 90° elbow, environmental (style B) – Continued.

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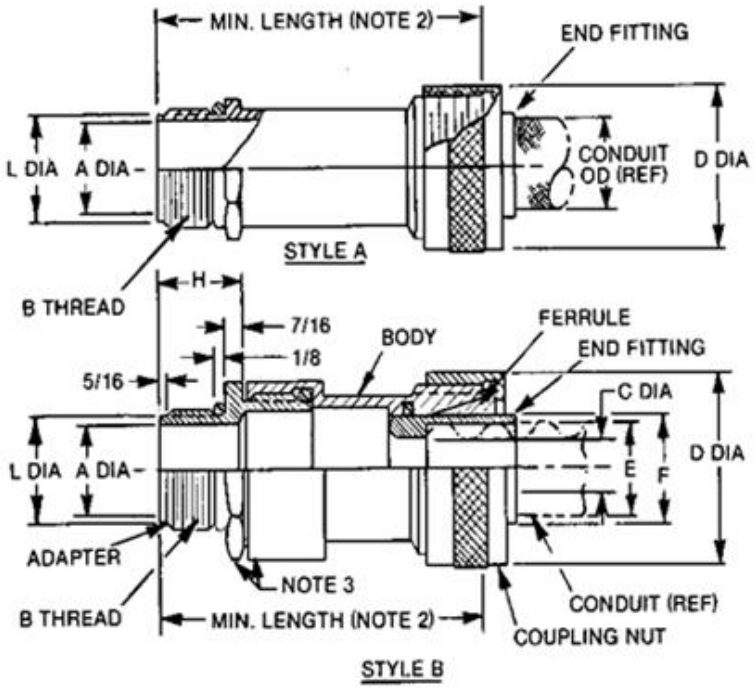


TABLE 5-1. GR2123 adapter sizes.

Adapter End				
Dash No.	**B Thread (Class 2A)**	**A Dia.**	**L Dia.**	**H Dim.**
1	1⅛ - 14 N	0.500	0.875	1 9/16
2	1⅛ - 12 UN	0.609	0.984	1 9/16
3	1.288 - 14 N	0.750	1.100	1 9/16
4	¾ - 14 NPSM	0.750	1.100	1 9/16
5	1 5/16 - 12 UN	0.812	1.171	1 9/16
6	1 - 11½ NPSM	1.000	1.170	1 9/16
7	1¼ - 11½ NPSM	1.000	1.400	1 9/16
8	1⅝ - 12 UN	1.062	1.484	1 9/16
9	1¼ - 11½ NPSM	1.180	1.500	1 9/16
10	2.025 - 11½ N	1.250	1.650	1⅞
11	1 13/16 - 12 UN	1.265	1.671	1⅞
12	2 1/16 - 12 UN	1.437	1.921	2
13	2¼ - 11½ N	1.500	1.930	2
14	2 5/16 - 12 UN	1.625	2.173	2⅛
15	2 11/16 - 12 UN	1.875	2.531	2 5/16
16	2½ - 8 NPSM	2.000	2.600	2 5/16
17	3 - 12 UN	2.187	2.850	2 5/16
18	3⅛ - 12 UN	2.312	2.968	2⅜
19	3 - 8 NPSM	2.500	3.000	2⅜

FIGURE 5. GR2123 transition fitting for bulkhead stuffing tube.

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TABLE 5-2. GR2123 transition sizes.

Standard Size Conduit			Transition Fitting				
			End Fitting			Nut	
Conduit Part Number	Conduit Size (Nom. ID)	OD (Over Braid)	C Dia.	E +0.010 -0.000	F ±0.005	D Max.	Dash No.
EM02	¼	0.42±0.01	0.25	0.450	0.530	1.03	EM02
EM03	⅜	0.54±0.01	0.38	0.570	0.650	1.23	EM03
EM04	½	0.69±0.01	0.50	0.720	0.800	1.36	EM04
EM05	⅝	0.82±0.01	0.62	0.850	0.930	1.59	EM05
EM06	¾	0.94±0.01	0.75	0.970	1.050	1.73	EM06
EM08	1	1.25±0.01	1.00	1.280	1.360	1.91	EM08
EM10	1¼	1.50±0.01	1.25	1.530	1.610	2.28	EM10
EM12	1½	1.75±0.01	1.50	1.780	1.860	2.66	EM12
EM16	2	2.28±0.015	2.00	2.315	2.395	3.17	EM16
EM20	2½	2.78±0.015	2.50	2.815	2.895	3.97	EM20
EM24	3	3.28±0.015	3.00	3.315	3.395	4.35	EM24

GR2123 PIN Construction:

Example:	**M32771-GR2123**	-	**2**	**3**	**EM04**
	Basic PIN	-	Dash No. (see table 5-1)	Body/Adapter Length (see note 2)	Dash No. (see table 5-2)

NOTES:

1. Material shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404, except end fitting shall be CRES, 300 series, passivated.
2. Standard minimum length shall be 3 inches for one-piece (style A) and 3.5 inches for two-piece (style B).
3. When conduit OD exceeds “A” diameter, a two-piece (style B) shall be supplied.
4. Assembly identified in accordance with MIL-STD-130.
5. Discard stuffing-tube gland nut.
6. Basic PIN without O-rings is M32771-GR2123; basic PIN with O-rings is M32771-GRO2123.
7. Dimensions are in inches.
8. The American National thread series (N) has been superseded by the United thread series (UN). The “N” thread sizes listed in table 5-1 are retained for use with those items that have “N” threads.

FIGURE 5. GR2123 transition fitting for bulkhead stuffing tube – Continued.

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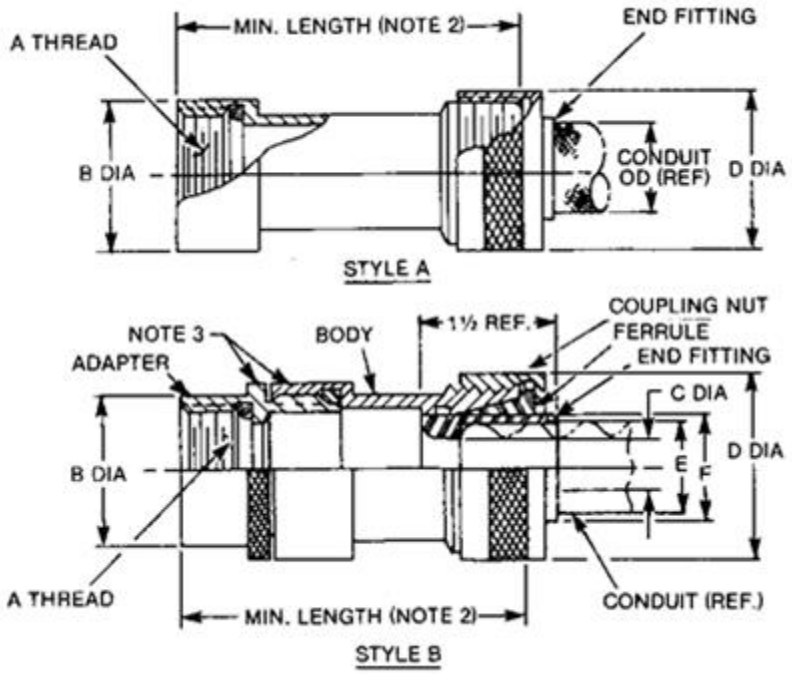


TABLE 6-1. GR2124 adapter sizes.

Backshell		Adapter End	
Size	**Dash No.**	**A Thread Class 2B**	**B Dia. +0.010 -0.020**
8S, 10S	3	½ - 28 UNEF	0.625
10SL, 12, 12S	4	⅝ - 24 UNEF	0.750
12SL, 14, 14S	6	¾ - 20 UNEF	0.875
16, 16S	8	⅞ - 20 UNEF	1.000
18	10	1 - 20 UNEF	1.125
20, 22	12	1 3/16 - 18 UNEF	1.312
24, 28	16	1 7/16 - 18 UNEF	1.562
32	20	1¾ - 18 UNS	1.875
36	24	2 - 18 UNS	2.125
40	28	2¼ - 16 UN	2.375
44	32	2½ - 16 UN	2.625
48	40	3 - 16 UN	3.125

FIGURE 6. GR2124 transition fitting for MS-series backshell.

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Check the source to verify that this is the current version before use.

TABLE 6-2. GR2123 transition sizes.

Standard Size Conduit			Transition Fitting				
			End Fitting			Nut	
Conduit Part Number	Conduit Size (Nom. ID)	OD (Over Braid)	C Dia.	E +0.010 -0.000	F ±0.005	D Max.	Dash No.
EM02	¼	0.42±0.01	0.25	0.450	0.530	1.03	EM02
EM03	⅜	0.54±0.01	0.38	0.570	0.650	1.23	EM03
EM04	½	0.69±0.01	0.50	0.720	0.800	1.36	EM04
EM05	⅝	0.82±0.01	0.62	0.850	0.930	1.59	EM05
EM06	¾	0.94±0.01	0.75	0.970	1.050	1.73	EM06
EM08	1	1.25±0.01	1.00	1.280	1.360	1.91	EM08
EM10	1¼	1.50±0.01	1.25	1.530	1.610	2.28	EM10
EM12	1½	1.75±0.01	1.50	1.780	1.860	2.66	EM12
EM16	2	2.28±0.015	2.00	2.315	2.395	3.17	EM16
EM20	2½	2.78±0.015	2.50	2.815	2.895	3.97	EM20
EM24	3	3.28±0.015	3.00	3.315	3.395	4.35	EM24

GR2124 PIN Construction:

Example:	**M32771-GR2124**	-	**3**	**EM02**	**2.5**
	Basic PIN	-	Dash No. (see table 6-1)	Dash No. (see table 6-2)	Body/Adapter Length (see note 2)

NOTES:
1. Material shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404, except end fitting shall be CRES, 300 series, passivated.
2. Standard minimum length shall be 2 inches for one-piece (style A) and 2.5 inches for two-piece (style B).
3. When conduit OD exceeds ID of connector shell, a two-piece (style B) shall be supplied.
4. Assembly identified per MIL-STD-130.
5. Basic PIN without O-rings is M32771-GR2124; basic PIN with O-rings is M32771-GRO2124.
6. Dimensions are in inches.

FIGURE 6. GR2124 transition fitting for MS-series backshell – Continued.

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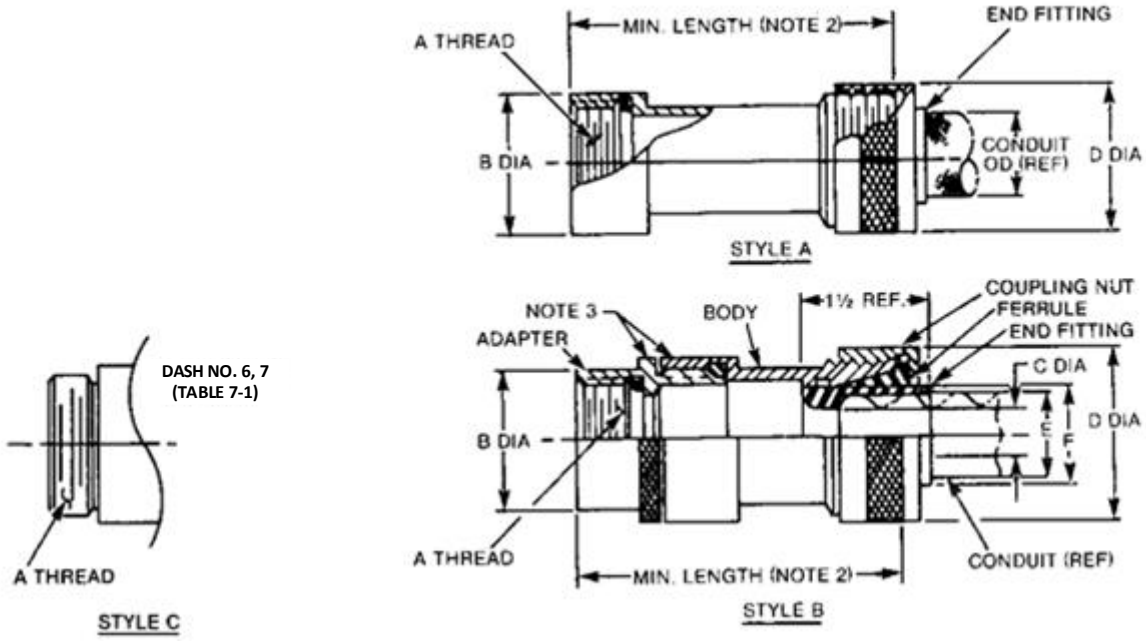


TABLE 7-1. GR2125 adaptor reference.

Adapter End			Connector Part No.		Cable Type	Conduit Size (Nom)
Dash No.	A Thread	B Dia.	ITT/CANNON	(Formerly) GREMAR		
4	¾ - 16 UNF-2B	0.875	152105-2380	5633	RG-223/U RG-58C/U	¼
5	⅞ - 20 UNEF-2B	1.062	152100-2390	7991	RG-214A/U RG-98/U	½
6	1 1/16 - 18 UNEF-2A	1.187	152190-0000	16312	RG-14A/U RG-293A/U	¾
7	1 1/8 - 18 UNEF-2A	1.187	---	11865	RG-17A/U	1

FIGURE 7. GR2125 transition fitting for triaxial connector.

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TABLE 7-2. GR2125 sizes.

Standard Size Conduit			Transition Fitting				
Conduit Part Number	Conduit Size (Nom. ID)	OD (Over Braid)	End Fitting			Nut	Dash No.
			C Dia.	E +0.010 -0.000	F ±0.005	D Max.	
EM02	¼	0.42±0.01	0.25	0.450	0.530	1.03	EM02
EM03	⅜	0.54±0.01	0.38	0.570	0.650	1.23	EM03
EM04	½	0.69±0.01	0.50	0.720	0.800	1.36	EM04
EM05	⅝	0.82±0.01	0.62	0.850	0.930	1.59	EM05
EM06	¾	0.94±0.01	0.75	0.970	1.050	1.73	EM06
EM08	1	1.25±0.01	1.00	1.280	1.360	1.91	EM08
EM10	1¼	1.50±0.01	1.25	1.530	1.610	2.28	EM10
EM12	1½	1.75±0.01	1.50	1.780	1.860	2.66	EM12
EM16	2	2.28±0.015	2.00	2.315	2.395	3.17	EM16
EM20	2½	2.78±0.015	2.50	2.815	2.895	3.97	EM20
EM24	3	3.28±0.015	3.00	3.315	3.395	4.35	EM24

GR2125 PIN Construction:

Example:	**M32771-GR2125**	-	**4**	**EM02**	**2.5**
	Basic PIN	-	Dash No. (see table 7-1)	Dash No. (see table 7-2)	Body/Adapter Length (see note 2)

NOTES:
1. Material shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404, except end fitting shall be CRES, 300 series, passivated.
2. Standard minimum length shall be 2 inches for one-piece (style A) and 2.5 inches for two-piece (style B).
3. When conduit OD exceeds ID of connector shell, a two-piece (style B or C) shall be supplied.
4. Assembly identified per MIL-STD-130.
5. Basic PIN without O-rings is M32771-GR2125; basic PIN with O-rings is M32771-GRO2125
6. Dimensions are in inches.

FIGURE 7. GR2125 transition fitting for triaxial connector – Continued.

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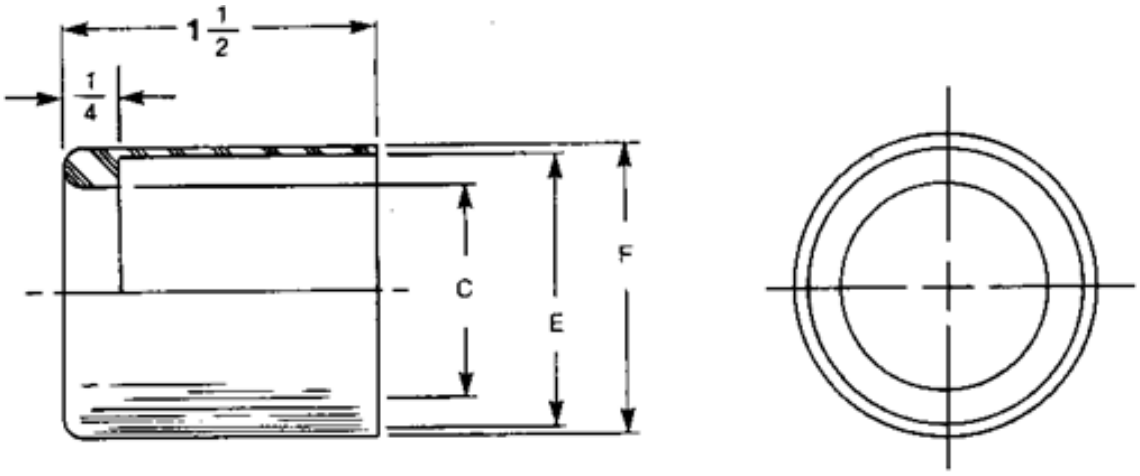


TABLE 8-1. GR2126 sizes.

Standard Size Conduit			End Fitting			
Conduit Part Number	**Conduit Size (Nom. ID)**	**OD (Over Braid)**	**C Dia.**	**E +0.010 -0.000**	**F ±0.005**	**End Fitting PIN**
EM02	¼	0.42±0.01	0.25	0.450	0.530	M32771-GR2126-EM02
EM03	⅜	0.54±0.01	0.38	0.570	0.650	M32771-GR2126-EM03
EM04	½	0.69±0.01	0.50	0.720	0.800	M32771-GR2126-EM04
EM05	⅝	0.82±0.01	0.62	0.850	0.930	M32771-GR2126-EM05
EM06	¾	0.94±0.01	0.75	0.970	1.050	M32771-GR2126-EM06
EM08	1	1.25±0.01	1.00	1.280	1.360	M32771-GR2126-EM08
EM10	1¼	1.50±0.01	1.25	1.530	1.610	M32771-GR2126-EM10
EM12	1½	1.75±0.01	1.50	1.780	1.860	M32771-GR2126-EM12
EM16	2	2.28±0.015	2.00	2.315	2.395	M32771-GR2126-EM16
EM20	2½	2.78±0.015	2.50	2.815	2.895	M32771-GR2126-EM20
EM24	3	3.28±0.015	3.00	3.315	3.395	M32771-GR2126-EM24

NOTES:
1. Material shall be CRES, 300 series, passivated.
2. Identify in accordance with MIL-STD-130.
3. Dimensions are in inches.

FIGURE 8. GR2126 end fitting for flexible conduit.

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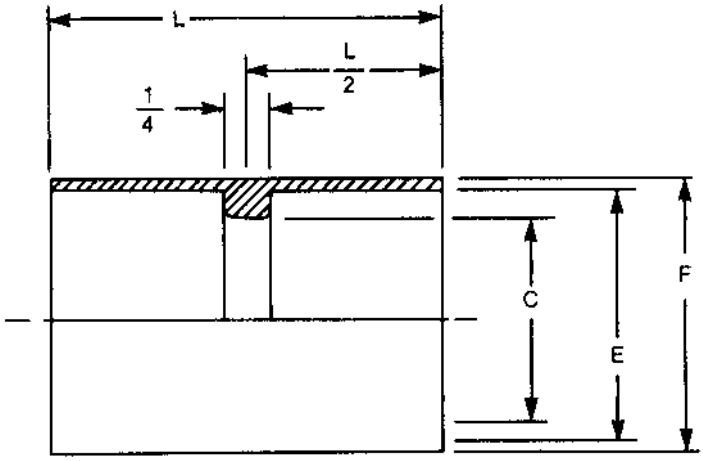


TABLE 9-1. GR2127 sizes.

Standard Size Conduit			Conduit Coupling			
Conduit Part Number	**Conduit Size (Nom. ID)**	**OD (Over Braid)**	**C Dia.**	**E +0.010 -0.000**	**F ±0.005**	**L ±0.093**
EM02	¼	0.42±0.01	0.25	0.450	0.530	1.25
EM03	⅜	0.54±0.01	0.38	0.570	0.650	1.37
EM04	½	0.69±0.01	0.50	0.720	0.800	1.50
EM05	⅝	0.82±0.01	0.62	0.850	0.930	1.62
EM06	¾	0.94±0.01	0.75	0.970	1.050	1.75
EM08	1	1.25±0.01	1.00	1.280	1.360	2.00
EM10	1¼	1.50±0.01	1.25	1.530	1.610	2.12
EM12	1½	1.75±0.01	1.50	1.780	1.860	2.25
EM16	2	2.28±0.015	2.00	2.315	2.395	2.50
EM20	2½	2.78±0.015	2.50	2.815	2.895	2.75
EM24	3	3.28±0.015	3.00	3.315	3.395	3.00

GR2127 PIN Construction:

Example: **M32771-GR2127** - **EM06**

Basic PIN - Conduit Part No. (see table 9-1)

NOTES:
1. Material shall be low-carbon steel with cadmium plating in accordance with SAE AMS-QQ-P-416.
2. Identify in accordance with MIL-STD-130.
3. Dimensions are in inches.

FIGURE 9. GR2127 conduit coupling.

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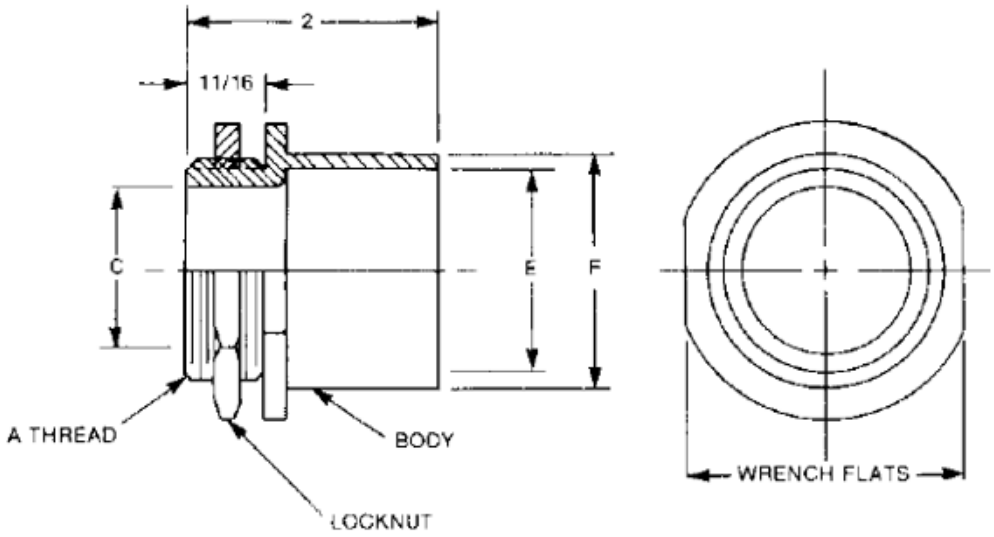


TABLE 10-1. GR2128 sizes.

Standard Conduit			Panel Fitting				
Conduit Part Number	**Conduit Size (Nom. ID)**	**OD (Over Braid)**	**A Thread Class 2A**	**C Dia.**	**E +0.010 -0.000**	**F ±0.005**	**Panel Fitting PIN**
EM02	¼	0.42±0.01	0.625 - 24 UNEF	0.25	0.450	0.530	M32771-GR2128-EM02
EM03	⅜	0.54±0.01	0.750 - 20 UNEF	0.38	0.570	0.650	M32771-GR2128-EM03
EM04	½	0.69±0.01	0.875 - 20 UNEF	0.50	0.720	0.800	M32771-GR2128-EM04
EM05	⅝	0.82±0.01	1.000 - 20 UNEF	0.62	0.850	0.930	M32771-GR2128-EM05
EM06	¾	0.94±0.01	1.1875 - 18 UNEF	0.75	0.970	1.050	M32771-GR2128-EM06
EM08	1	1.25±0.01	1.4375 - 18 UNEF	1.00	1.280	1.360	M32771-GR2128-EM08
EM10	1¼	1.50±0.01	1.750 - 18 UNS	1.25	1.530	1.610	M32771-GR2128-EM10
EM12	1½	1.75±0.01	2.000 - 18 UNS	1.50	1.780	1.860	M32771-GR2128-EM12
EM16	2	2.28±0.015	2.500 - 16 UN	2.00	2.315	2.395	M32771-GR2128-EM16
EM20	2½	2.78±0.015	3.000 - 16 UNS	2.50	2.815	2.895	M32771-GR2128-EM20
EM24	3	3.28±0.015	3.500 - 16 UN	3.00	3.315	3.395	M32771-GR2128-EM24

NOTES:
1. Body material shall be CRES, 300 series, passivated.
2. Locknut material shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404.
3. Identify in accordance with MIL-STD-130.
4. Dimensions are in inches.

FIGURE 10. GR2128 panel fitting.

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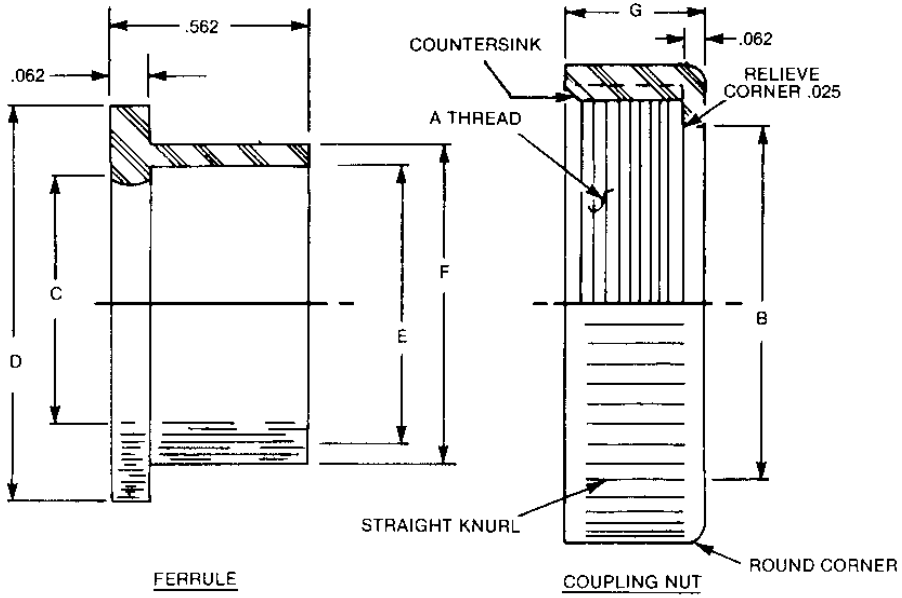


TABLE 11-1. GR2129 sizes.

Standard Size Conduit			Ferrule and Coupling Nut Assembly							
			Ferrule				Coupling Nut			
Conduit Part Number	Conduit Size (Nom. ID)	OD (Over Braid)	C Dia.	E +0.010 -0.000	F ±0.005	D ±0.010	A Thread Class 2B	B ±0.005	G ±0.015	Assembly PIN
EM02	¼	0.42±0.01	0.25	0.450	0.530	0.635	0.750 - 20 UNEF	0.545	0.438	M32771-GR2129-EM02
EM03	⅜	0.54±0.01	0.38	0.570	0.650	0.760	0.875 - 20 UNEF	0.665	0.438	M32771-GR2129-EM03
EM04	½	0.69±0.01	0.50	0.720	0.800	0.906	1.000 - 20 UNEF	0.815	0.438	M32771-GR2129-EM04
EM05	⅝	0.82±0.01	0.62	0.850	0.930	1.072	1.188 - 18 UNEF	0.945	0.438	M32771-GR2129-EM05
EM06	¾	0.94±0.01	0.75	0.970	1.050	1.322	1.438 - 18 UNEF	1.065	0.438	M32771-GR2129-EM06
EM08	1	1.25±0.01	1.00	1.280	1.360	1.635	1.750 - 18 NS	1.375	0.500	M32771-GR2129-EM08
EM10	1¼	1.50±0.01	1.25	1.530	1.610	1.885	2.000 - 18 NS	1.625	0.562	M32771-GR2129-EM10

FIGURE 11. GR2129 ferrule and coupling nut for flexible circuit.

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TABLE 11-1. GR2129 sizes - Continued.

Standard Size Conduit			Ferrule and Coupling Nut Assembly							
			Ferrule				Coupling Nut			
Conduit Part Number	Conduit Size (Nom. ID)	OD (Over Braid)	C Dia.	E +0.010 -0.000	F ±0.005	D ±0.010	A Thread Class 2B	B ±0.005	G ±0.015	Assembly PIN
EM12	1½	1.75±0.01	1.50	1.780	1.860	2.135	2.250 - 16 UN	1.875	0.562	M32771-GR2129-EM12
EM16	2	2.28±0.015	2.00	2.315	2.395	2.885	3.000 - 16 UN	2.410	0.688	M32771-GR2129-EM16
EM20	2½	2.78±0.015	2.50	2.815	2.895	3.260	3.375 - 16 UN	2.910	0.688	M32771-GR2129-EM20
EM24	3	3.28±0.015	3.00	3.315	3.395	3.760	3.875 - 16 UN	3.410	0.750	M32771-GR2129-EM24

NOTES:

1. Ferrule material shall be CRES, 300 series, passivated.
2. Coupling nut shall be low-carbon steel with electroless nickel coating in accordance with SAE AMS2404.
3. The “A” threads shall be perfect to within 0.094 inch from bottom. No thread relief at inner end permitted.
4. Identify in accordance with MIL-STD-130.
5. Dimensions are in inches.
6. The American National thread series (N) has been superseded by the United thread series (UN). The “NS” thread sizes listed are retained for use with those items that have “NS” threads.
7. The M32771-GR2129-EM06 assembly couples ¾-inch conduit to the stuffing-tube adapter, S9407-AB-HBK-010.

FIGURE 11. GR2129 ferrule and coupling nut for flexible circuit - Continued.

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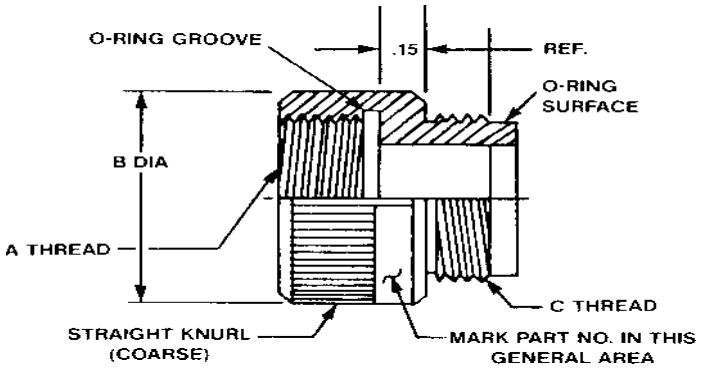


TABLE 12-1. GR2130 cross reference.

Thread Size Code	Mates with GR2000-Series							A Thread Class 2B	B Dia. ±0.02	C Thread Class 2A
	2120	2121	2122	2123	2124	2125	2129			
01	√		√					⅜ - 32 UNEF	0.69	⅜ - 32 UNEF
02	√		√					7/16 - 27 UNS	0.75	7/16 - 27 UNS
03	√		√					7/16 - 28 UNS	0.75	7/16 - 28 UNS
04	√	√	√		√			½ - 28 UNEF	0.82	½ - 28 UNEF
05	√		√					9/16 - 24 UNEF	0.88	9/16 - 24 UNEF
06	√	√	√		√			⅝ - 24 UNEF	0.94	⅝ - 24 UNEF
07	√		√					11/16 - 24 UNEF	1.00	11/16 - 24 UNEF
08						√		¾ - 16 UNEF	1.06	¾ - 16 UNEF
09	√	√	√		√		√	¾ - 20 UNEF	1.06	¾ - 20 UNEF
10	√	√	√		√	√	√	⅞ - 20 UNEF	1.19	⅞ - 20 UNEF
11	√	√	√		√		√	1 - 20 UNEF	1.31	1 - 20 UNEF
12						√		1 1/16 - 18 UNEF	1.38	1 1/16 - 18 UNEF
13				√				1⅛ - 12 UN	1.44	1⅛ - 12 UN
14				√				1⅛ - 14 N	1.44	1⅛- 14 N
15	√		√					1⅛ - 18 UNEF	1.44	1 1/18 - 18 UNEF
16	√		√					1⅛ - 24 UNS	1.44	1 1/18 - 24 UNS
17		√			√		√	1 3/16 - 18 UNEF	1.50	1 3/16 - 18 UNEF
18	√		√					1¼ - 18 UNEF	1.56	1¼ - 18 UNEF
19				√				1-288 - 14 UNEF	1.60	1-288 - 14 UNEF
20				√				1 5/16 - 12 UN	1.63	1 5/16 - 12 UN
21	√		√					1⅜ - 18 UNEF	1.69	1⅜ - 18 UNEF
22		√			√		√	1 7/16 - 18 UNEF	1.75	1 7/16 - 18 UNEF
23				√				1⅝ - 12 UN	1.94	1⅝ - 12 UN
24	√		√					1⅝ - 18 UNEF	1.94	1⅝ - 18 UNEF
25		√			√		√	1¾ - 18 NS	2.06	1¾ - 18 NS
26				√				1 3/16 - 12 UN	2.13	1 3/16 - 12 UN
27	√		√					1⅞ - 16 UN	2.19	1-⅞ - 16 UN
28	√		√					1 29/32 - 18 UNS	2.22	11 29/32 - 18 UNS
29	√		√		√		√	2 - 18 NS	2.31	2 - 18 NS

FIGURE 12. GR2130 isolator adapter.

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TABLE 12-1. GR2130 cross reference - Continued.

Thread Size Code	Mates with GR2000-Series							A Thread Class 2B	B Dia. ±0.02	C Thread Class 2A
	2120	2121	2122	2123	2124	2125	2129			
30				√				2¼ - 11½ N	2.34	2¼ - 11½ N
31				√				2 1/16 - 12 UN	2.38	2 1/16 - 12 UN
32	√		√					2 1/16 - 16 UN	2.38	2 1/16 - 16 UN
33	√		√	√				2 1/16 - 20 UNEF	2.38	2 1/16 - 20 UNEF
34	√		√					2⅛-16 UN	2.44	2⅛ - 16 UN
35	√		√					2⅛ - 18 UNS	2.44	2⅛ - 18 UNS
36				√				2¼ - 11½ N	2.56	2¼ - 11½ N
37		√			√			2¼ - 16 UN	2.56	2¼- 16 UN
38				√				2 5/16 - 12 UN	2.63	2 5/16 - 12 UN
39	√		√					2 5/16 - 16 UN	2.63	2 5/16 - 16 UN
40	√		√				√	2⅜ - 16 UN	2.69	2⅜ - 16 UN
41		√			√		√	2½ - 16 UN	2.81	2½ - 16 UN
42	√		√				√	2⅝ - 16 UN	2.94	2⅝ - 16 UN
43				√				2 11/16 - 12 UN	3.00	2 11/16 - 12 UN
44	√		√					2 13/16 - 18 UNS	3.13	2 13/16 - 18 UNS
45	√		√					2⅞ - 16 UN	3.19	2⅞ - 16 UN
46				√				3 - 12 UN	3.31	3 - 12 UN
47		√			√			3 - 16 UN	3.31	3 - 16 UN
48				√			√	3⅛ - ½ UN	3.44	3⅛ - ½ UN
49								3⅜ - 16 UN	3.69	3⅜ - 16 UN
50								3⅞ - 16 UN	4.19	3⅞ - 16 UN
51				√				¾ - 14 NPSM	1.35	¾ - 14 NPSM
52				√				1 - 11½ NPSM	1.61	1 - 11½ NPSM
53				√				1¼- 11½ NPSM	1.95	1¼ - 11½ NPSM
54				√				2½ - 8 NPSM	3.15	2½ - 8 NPSM
55				√				3 - 8 NPSM	3.78	3 - 8 NPSM

TABLE 12-2. GR2130 material reference.

Material Code	Material	Specification
N	Nylon, rigid: rods, tubes, etc.	L-P-410
	Nylon, molding plastic	ASTM D4066
STN	Super tough Nylon Zytel, ST801 (or equal)	Dupont spec. sheets or equivalent source

FIGURE 12. GR2130 isolator adapter – Continued.

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GR2130 PIN Construction:

Example:	**M32771-GR2130**	-	**05**	**N**	**10SL**
	Basic PIN	-	Thread Size (see table 12-1)	Material Code (see table 12-2)	Connector Shell Size (see table 12-1 of GR2120, 2121, 2122, or 2124) (leave blank for GR2123, 2125, or 2129)

NOTES:

1. The GR2130 isolator adapter is made of a non-conductive material to electrically insulate the GR2000-series fittings from the associated connectors, backshells, stuffing-tubes, etc.
2. Basic PIN without O-rings is M32771-GR2130; basic PIN with O-rings is M32771-GRO2130.
3. M32771-GR2130-51 through M32771-GR2130-55 adapters have NPSM threads (straight pipe threads for mechanical joints).
4. Parts identified in accordance with MIL-STD-130.
5. Dimensions are in inches.

FIGURE 12. GR2130 isolator adapter – Continued.

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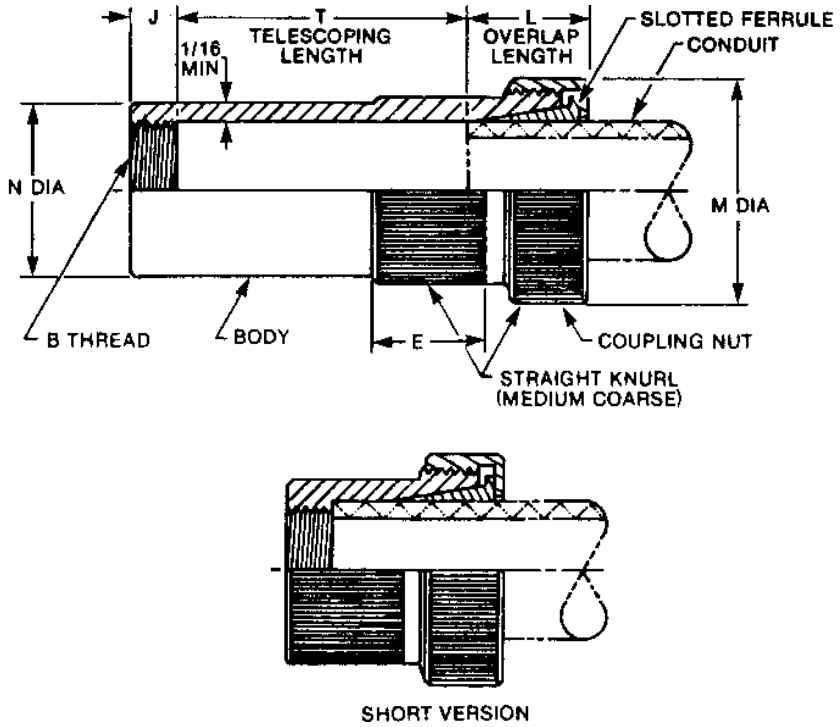


TABLE 13-1. RP2100 sizes.

Standard Conduit		End Fitting						
Conduit Part Number	**Conduit Size (Nom. ID)**	**B Thread (Class 2B)**	**J Thread Length +0.06 -0.0**	**T Telescoping Length ±0.12**	**L Overlap Length ±0.06**	**E Knurl Length ±0.06**	**M OD (Max.)**	**N Dia. (Max.)**
EM02	¼	0.4375 – 28 UNEF	0.21	1.50	0.75	0.50	0.93	0.70
EM03	⅜	0.5625 – 24 UNEF	0.25	1.50	0.75	0.50	1.12	0.82
EM04	½	0.6875 – 24 UNEF	0.25	1.75	0.75	0.62	1.25	0.97
EM05	⅝	0.8125 – 20 UNEF	0.31	1.75	0.75	0.62	1.43	1.12
EM06	¾	0.9375 – 20 UNEF	0.31	2.00	0.75	0.75	1.62	1.22
EM08	1	1.250 – 18 UNEF	0.37	2.00	1.00	0.75	1.81	1.53
EM10	1¼	1.5625 – 18 UNEF	0.37	2.50	1.00	0.87	2.18	1.81
EM12	1½	1.875 – 16 UN	0.43	2.50	1.00	0.87	2.56	2.12
EM16	2	2.375 – 16 UN	0.43	3.00	1.00	1.00	3.06	2.62
EM20	2½	2.875 – 16 UN	0.43	3.00	1.00	1.00	3.86	3.12
EM24	3	3.375 – 16 UN	0.43	3.00	1.00	1.00	4.38	3.62

FIGURE 13. RP2100 conduit end fitting, straight.

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RP2100 PIN Construction:

Example:	**M32771-RP2100**	-	**EM06**	**T or S**
	Basic PIN	-	Conduit Part No. (see table 13-1)	Length Code T – Telescoping Length S – Short Length (where telescoping feature not required)

NOTES:
1. The function of the RP2100 end fitting is to terminate flexible shielding conduit and provide a standard thread for attaching all RP2000-series adapters.
2. Dimensions are in inches.

FIGURE 13 RP2100 conduit end fitting, straight – Continued

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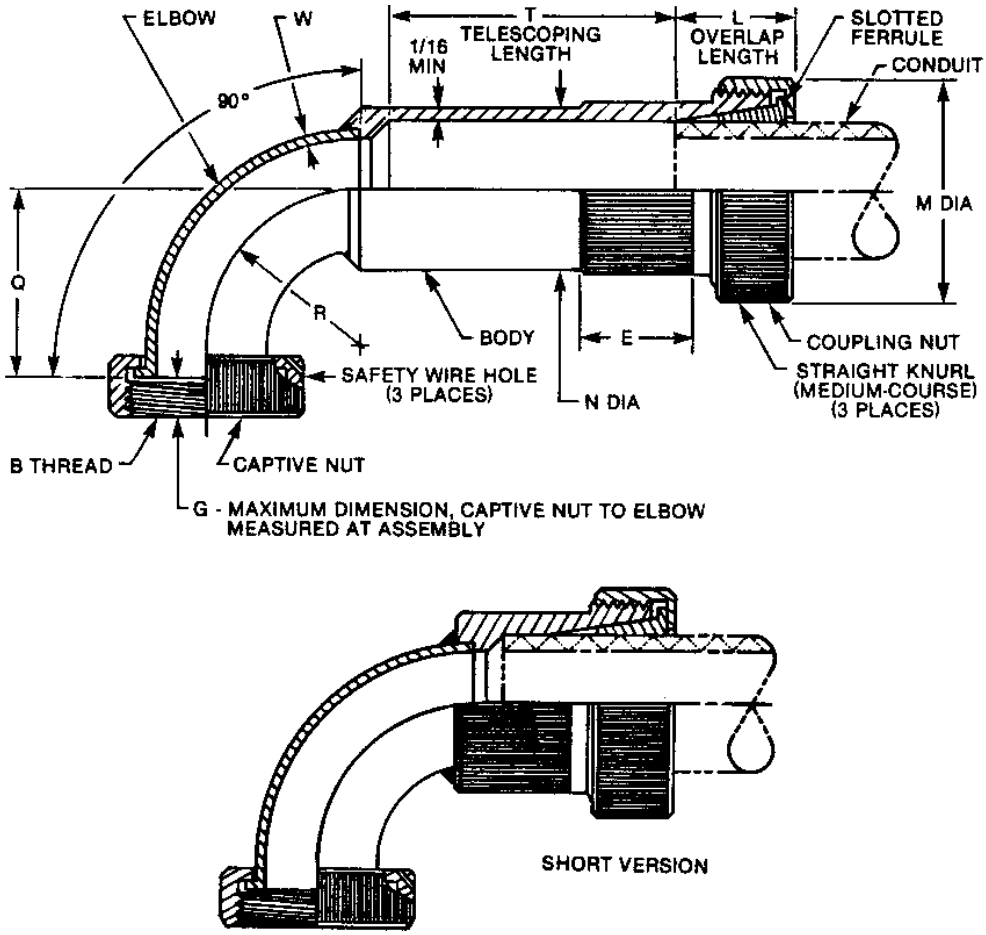


FIGURE 14. RP2110 conduit end fitting, 90°.

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TABLE 14-1. RP2110 sizes.

Standard Conduit		End Fitting									
Conduit Part Number	**Conduit Size (Nom. ID)**	**B Thread (Class 2B)**	**G +0.01 -0.02**	**Q ±0.06**	**R Bend Radius ±0.06**	**W Wall Thick. (Min.)**	**T TLSCPG Length ±0.12**	**L Overlap Length ±0.06**	**E Knurl Length ±0.06**	**M OD (Max.)**	**N Dia. (Max.)**
EM02	¼	0.4375 – 28 UNEF	0.14	0.81	0.43	0.028	1.50	0.75	0.50	0.93	0.70
EM03	⅜	0.5625 – 24 UNEF	0.17	0.93	0.56	0.028	1.50	0.75	0.50	1.12	0.82
EM04	½	0.6875 – 24 UNEF	0.17	1.12	0.75	0.028	1.75	0.75	0.62	1.25	0.97
EM05	⅝	0.8125 – 20 UNEF	0.23	1.25	0.87	0.028	1.75	0.75	0.62	1.43	1.12
EM06	¾	0.9375 – 20 UNEF	0.23	1.47	1.06	0.028	2.00	0.75	0.75	1.62	1.22
EM08	1	1.250 – 18 UNEF	0.29	1.81	1.37	0.028	2.00	1.00	0.75	1.81	1.53
EM10	1¼	1.5625 – 18 UNEF	0.29	2.12	1.68	0.028	2.50	1.00	0.87	2.18	1.81
EM12	1½	1.875 – 16 UN	0.35	2.50	2.06	0.058	2.50	1.00	0.87	2.56	2.12
EM16	2	2.375 – 16 UN	0.35	3.12	2.68	0.058	3.00	1.00	1.00	3.06	2.62
EM20	2½	2.875 – 16 UN	0.35	3.75	3.31	0.058	3.00	1.00	1.00	3.68	3.12
EM24	3	3.375 – 16 UN	0.35	5.00	4.50	0.058	3.00	1.00	1.00	4.38	3.62

RP2110 PIN Construction:

Example:	**M32771-RP2110**	-	**EM08**	**T or S**
	Basic PIN	-	Conduit Part No. (see table 14-1)	Length Code T – Telescoping Length S – Short Length (where telescoping feature not required)

NOTES:
1. The function of the RP2110 90° end fitting is to terminate flexible shielding conduit and provide a standard thread for attaching all RP2000-series adapters.
2. Dimensions are in inches.

FIGURE 14 RP2110 conduit end fitting, 90° – Continued.

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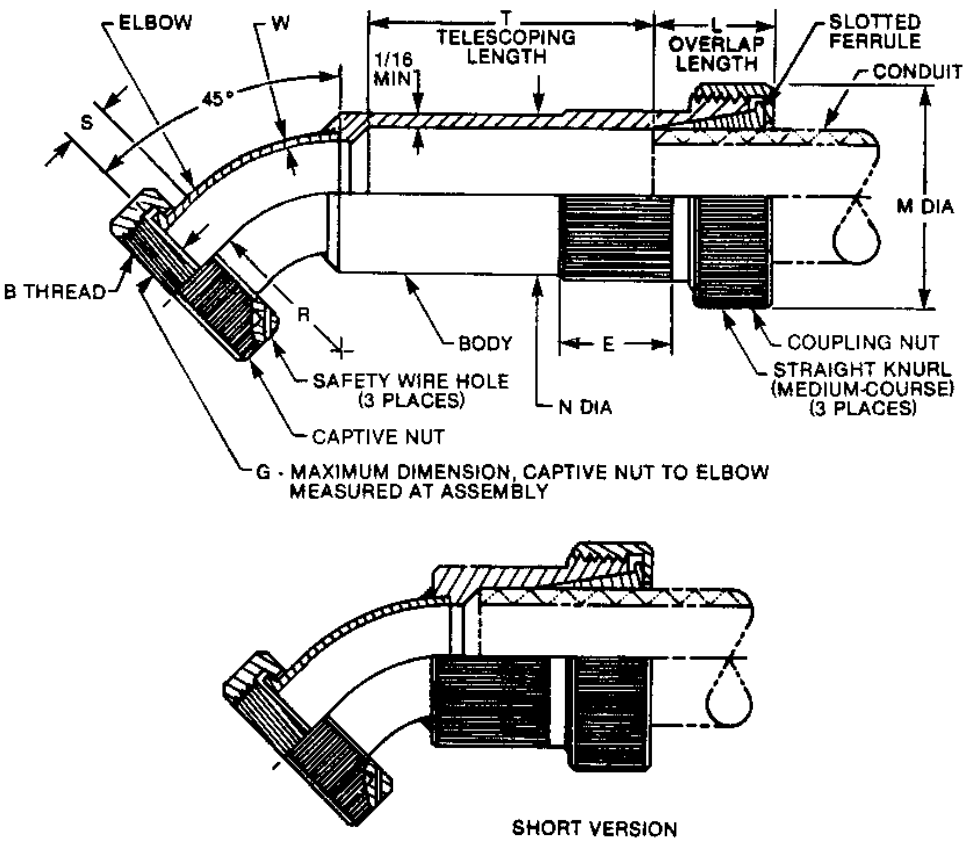


FIGURE 15. RP2120 conduit end fitting, 45°.

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TABLE 15-1. RP2120 sizes.

Standard Conduit		End Fitting									
Conduit Part Number	**Conduit Size (Nom. ID)**	**B Thread (Class 2B)**	**G +0.01 -0.02**	**S ±0.06**	**R Bend Radius ±0.06**	**W Wall Thick. (Min.)**	**T TLSCPG Length ±0.12**	**L Overlap Length ±0.06**	**E Knurl Length ±0.06**	**M OD (Max.)**	**N Dia. (Max.)**
EM02	¼	0.4375 – 28 UNEF	0.14	0.37	0.43	0.028	1.50	0.75	0.50	0.93	0.70
EM03	⅜	0.5625 – 24 UNEF	0.17	0.37	0.56	0.028	1.50	0.75	0.50	1.12	0.82
EM04	½	0.6875 – 24 UNEF	0.17	0.37	0.75	0.028	1.75	0.75	0.62	1.25	0.97
EM05	⅝	0.8125 – 20 UNEF	0.23	0.37	0.87	0.028	1.75	0.75	0.62	1.43	1.12
EM06	¾	0.9375 – 20 UNEF	0.23	0.37	1.06	0.028	2.00	0.75	0.75	1.62	1.22
EM08	1	1.250 – 18 UNEF	0.29	0.43	1.37	0.028	2.00	1.00	0.75	1.81	1.53
EM10	1¼	1.5625 – 18 UNEF	0.29	0.43	1.68	0.028	2.50	1.00	0.87	2.18	1.81
EM12	1½	1.875 – 16 UN	0.35	0.43	2.06	0.058	2.50	1.00	0.87	2.56	2.12
EM16	2	2.375 – 16 UN	0.35	0.43	2.68	0.058	3.00	1.00	1.00	3.06	2.62
EM20	2½	2.875 – 16 UN	0.35	0.43	3.31	0.058	3.00	1.00	1.00	3.68	3.12
EM24	3	3.375 – 16 UN	0.35	0.50	4.50	0.058	3.00	1.00	1.00	4.38	3.62

RP2120 PIN Construction:

Example:	**M32771-RP2120**	-	**EM08**	**T or S**
	Basic PIN	-	Conduit Part No. (see table 15-1)	Length Code T – Telescoping Length S – Short Length (where telescoping feature not required)

NOTES:
1. The function of the RP2120 45° end fitting is to terminate flexible shielding conduit and provide a standard thread for attaching all RP2000-series adapters.
2. Dimensions are in inches.

FIGURE 15 RP2120 conduit end fitting, 45° – Continued.

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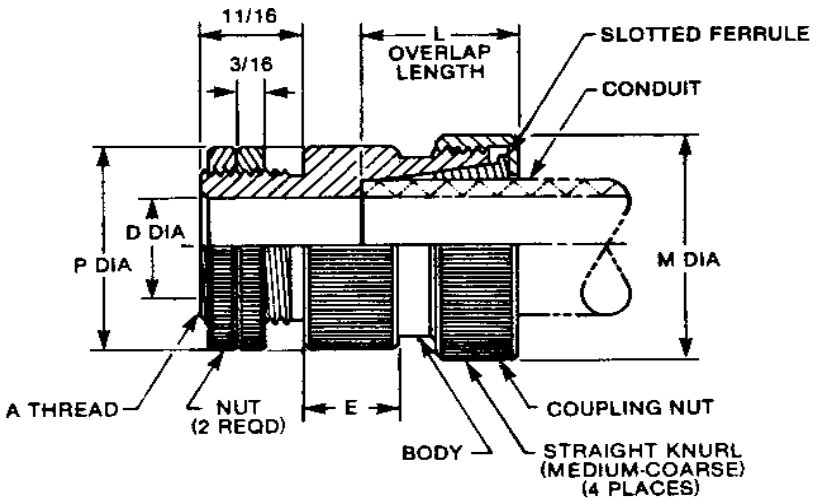


TABLE 16-1. RP2130 sizes.

Standard Conduit		End Fitting					
Conduit Part Number	**Conduit Size (Nom. ID)**	**A Thread (Class 2A)**	**D Dia. (Min.)**	**P Dia. (Min.)**	**L Overlap Length ±0.06**	**E Knurl Length ±0.06**	**M OD (Max.)**
EM02	¼	0.625 – 24 UNEF	0.25	0.87	0.75	0.50	0.93
EM03	⅜	0.750 – 20 UNEF	0.37	1.00	0.75	0.50	1.12
EM04	½	0.875 – 20 UNEF	0.50	1.12	0.75	0.56	1.25
EM05	⅝	1.000 – 20 UNEF	0.62	1.25	0.75	0.56	1.43
EM06	¾	1.1875 – 18 UNEF	0.75	1.43	0.75	0.62	1.62
EM08	1	1.4375 – 18 UNEF	1.00	1.68	1.00	0.62	1.81
EM10	1¼	1.750 – 16 UN	1.25	2.00	1.00	0.68	2.18
EM12	1½	2.00 – 16 UN	1.50	2.25	1.00	0.68	2.56
EM16	2	2.500 – 16 UN	2.00	3.00	1.00	0.75	3.06
EM20	2½	3.000 – 16 UN	2.50	3.50	1.00	0.75	3.50
EM24	3	3.500 – 16 UN	3.00	4.00	1.00	0.75	4.38

RP2130 PIN Construction:

Example: **M32771-RP2130** - **EM06**

Basic PIN - Conduit Part No. (see table 16-1)

NOTES:

1. The function of the RP2130 end fitting is to terminate flexible shielding conduit directly to a panel. Two nuts shall be supplied with this fitting. One nut secures the fitting to the panel, and a second nut acts as a lock nut to the first.
2. Dimensions are in inches.

FIGURE 16. RP2130 conduit end fitting, conduit-to-panel.

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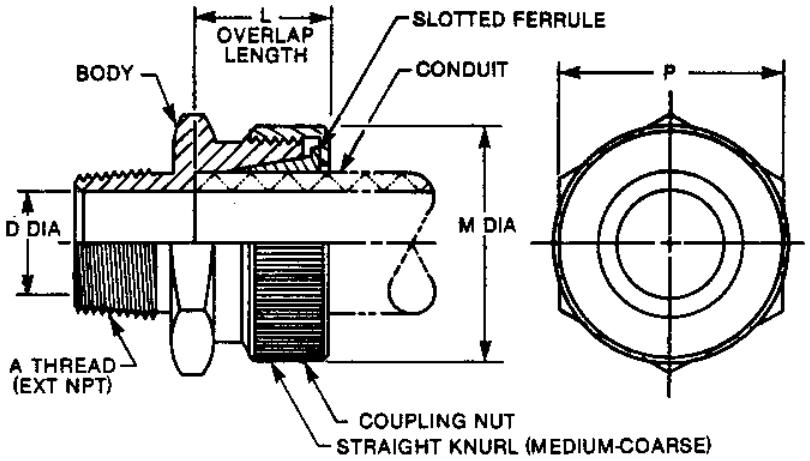


TABLE 17-1. RP2140 sizes.

Standard Conduit		End Fitting				
Conduit Part Number	**Conduit Size (Nom. ID)**	**A Pipe Thread (Class 2A)**	**D Dia. (Min.)**	**L Overlap Length ±0.06**	**M Max. OD Coupling Nut**	**P Min. Dim. Wrench Flats**
EM02	¼	¼ – 18 NPT	0.25	0.75	0.93	0.75
EM03	⅜	⅜ – 18 NPT	0.37	0.75	1.12	0.87
EM04	½	½ – 14 NPT	0.50	0.75	1.25	0.96
EM05	⅝	¾ – 14 NPT	0.62	0.75	1.43	1.37
EM06	¾	¾ – 14 NPT	0.75	0.75	1.62	1.18
EM08	1	1 – 11½ NPT	1.00	1.00	1.81	1.50
EM10	1¼	1¼ – 11½ NPT	1.25	1.00	2.18	1.81
EM12	1½	1½ – 11½ NPT	1.50	1.00	2.56	2.12
EM16	2	2 – 11½ NPT	2.00	1.00	3.06	2.62
EM20	2½	2½ – 8 NPT	2.50	1.00	3.50	3.25
EM24	3	3 – 8 NPT	3.00	1.00	4.38	3.87

RP2140 PIN Construction:

Example: **M32771-RP2140** - **EM06**

Basic PIN - Conduit Part No. (see table 17-1)

NOTES:
1. The function of the RP2140 end fitting is to terminate flexible shielding conduit with a reusable fitting having an external tapered pipe thread.
2. Dimensions are in inches.

FIGURE 17. RP2140 end fitting, conduit-to-pipe thread (tapered).

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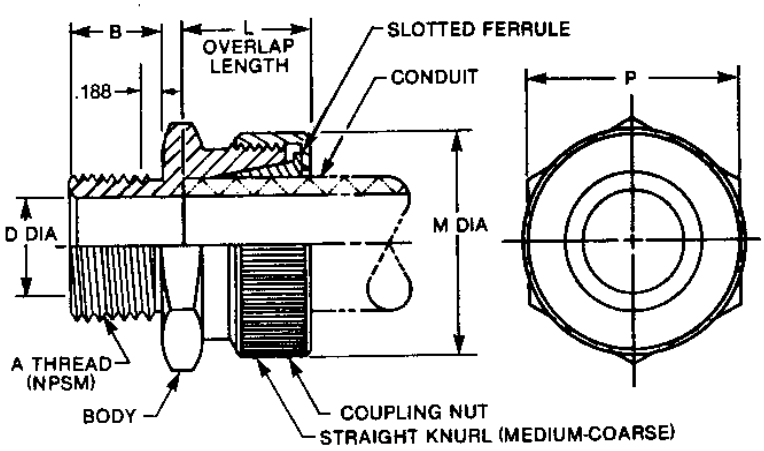


TABLE 18-1. RP2141 sizes.

Standard Conduit		End Fitting					
Conduit Part Number	**Conduit Size (Nom. ID)**	**A Pipe Thread (Class 2A)**	**B Dim. ±0.02**	**D Dia. (Min.)**	**L Overlap Length ±0.06**	**M Max. OD Coupling Nut**	**P Min Dim. Wrench Flats**
EM02	¼	¼ – 18 NPSM	0.75	0.25	0.75	0.93	0.75
EM03	⅜	⅜ – 18 NPSM	0.75	03.7	0.75	1.12	0.87
EM04	½	½ – 14 NPSM	0.87	0.50	0.75	1.25	0.96
EM05	⅝	¾ – 14 NPSM	0.87	0.62	0.75	1.43	1.37
EM06	¾	¾ – 14 NPSM	0.87	0.75	0.75	1.62	1.18
EM08	1	1 – 11½ NPSM	0.87	1.00	1.00	1.81	1.50
EM10	1¼	1¼ – 11½ NPSM	0.87	1.25	1.00	2.18	1.81
EM12	1½	1½ – 11½ NPSM	1.00	1.50	1.00	2.56	2.12
EM16	2	2 – 11½ NPSM	1.19	2.00	1.00	3.06	2.62
EM20	2½	2½ – 8 NPSM	1.62	2.50	1.00	3.50	3.25
EM24	3	3 – 8 NPSM	1.69	3.00	1.00	4.38	3.87

RP2141 PIN Construction:

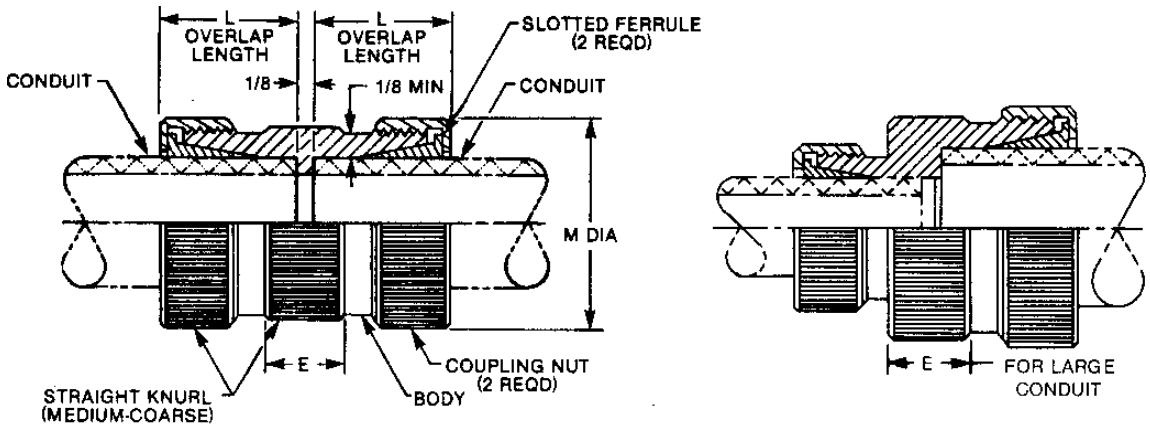
Example: **M32771-RP2141** - **EM06**

Basic PIN - Conduit Part No. (see table 18-1)

NOTES:
1. The function of the RP2141 end fitting is to terminate flexible shielding conduit with a reusable fitting having an external straight pipe thread.
2. Dimensions are in inches.

FIGURE 18. RP2141 end fitting, conduit-to-pipe thread (straight).

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STYLE 01
CONNECTS TWO IDENTICAL SIZES OF CONDUIT

STYLE 02
CONNECTS TWO DIFFERENT SIZES OF CONDUIT

TABLE 19-1. RP2200 sizes.

Standard Conduit		Coupling		
Conduit Part Number	**Conduit Size (Nom. ID)**	**L Overlap Length ±0.06**	**E Knurl Length ±0.06**	**M OD (Max.)**
EM02	¼	0.75	0.50	0.93
EM03	⅜	0.75	0.50	1.12
EM04	½	0.75	0.56	1.25
EM05	⅝	0.75	0.56	1.43
EM06	¾	0.75	0.62	1.62
EM08	1	1.00	0.62	1.81
EM10	1¼	1.00	0.68	2.18
EM12	1½	1.00	0.68	2.56
EM16	2	1.00	0.75	3.06
EM20	2½	1.00	0.75	3.50
EM24	3	1.00	0.75	4.38

FIGURE 19. RP2200 coupling, conduit-to-conduit.

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RP2200 Style 01 PIN Construction:

Example:	**M32771-RP2200**	-	**EM04**
	Basic PIN	-	Conduit Part No. (see table 19-1)

RP2200 Style 02 PIN Construction:

Example:	**M32771-RP2200**	-	**EM06**	**10**
	Basic PIN	-	Small Conduit Part No. (see table 19-1)	Large Conduit Part No. (see table 19-1)

NOTES:

1. The function of the RP2200 conduit-to-conduit coupling, style 01, is to connect two identical sizes of flexible shielding conduit; style 02 connects two different sizes of conduit.
2. For style 02 to provide a telescoping feature, the ID of the large conduit shall be larger than the OD of the small conduit.
3. Example of style 02 use: an inline connector (or similar component) installed in a cable can be shielded with the aid of these couplings. The connector is enclosed in a short length of large conduit, which is then coupled at each end to the small conduit that encloses the cable.
4. Dimensions are in inches.

FIGURE 19 RP2200 coupling, conduit-to-conduit – Continued.

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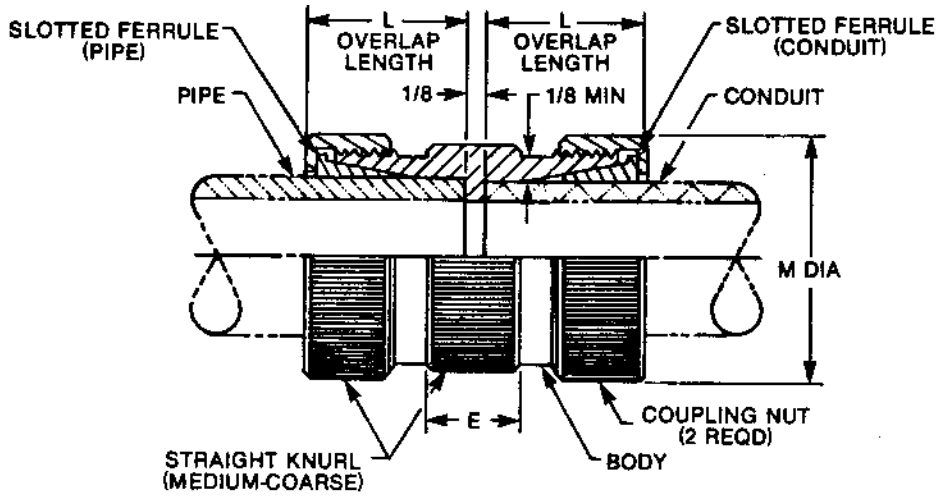


TABLE 20-1. RP2210 sizes.

Standard Conduit		Pipe (MIL-P-24691)			Coupling		
Conduit Part Number	Conduit Size (Nom. ID)	Pipe Size (Nom. ID)	Pipe OD (Ref.)		L Overlap Length ±0.06	E Knurl Length ±0.06	M OD (Max.)
			OD	Tolerance			
EM02	¼	¼	0.540	+0.015 – 0.031	0.75	0.50	0.93
EM03	⅜	⅜	0.675	+0.015 – 0.031	0.75	0.05	1.12
EM04	½	½	0.840	+0.015 – 0.031	0.75	0.56	1.25
EM06	¾	¾	1.050	+0.015 – 0.031	0.75	0.62	1.62
EM08	1	1	1.315	+0.015 – 0.031	1.00	0.62	1.81
EM10	1¼	1¼	1.660	+0.015 – 0.031	1.00	0.68	2.18
EM12	1½	1½	1.900	+0.015 – 0.031	1.00	0.68	2.56
EM16	2	2	2.375	+0.031 – 0.031	1.00	0.75	3.06
EM20	2½	2½	2.875	+0.031 – 0.031	1.00	0.75	3.50
EM24	3	3	3.500	+0.031 – 0.031	1.00	0.75	4.38

RP2210 PIN Construction:

Example: **M32771-RP2210** - **EM08**

Basic PIN - Conduit Part No. (see table 20-1)

NOTES:
1. The function of the RP2210 coupling is to attach flexible shielding conduit to rigid shielding conduit ("black iron" pipe).
2. Dimensions are in inches.

FIGURE 20. RP2210 coupling, conduit-to-pipe.

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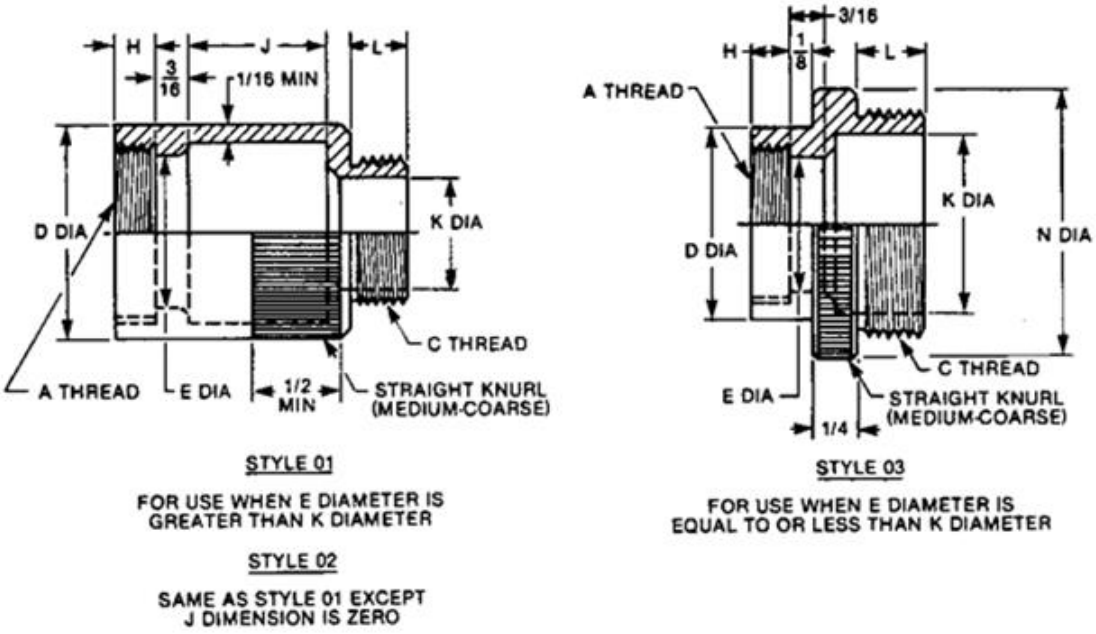


TABLE 21-1. RP2311 sizes.

Standard Conduit		Adapter			
Conduit Size Code	Conduit Size (Nom. ID)	C Thread (Class 2A)	L ±0.02	K Dia. (Min.)	N Dia. ±0.02
02	¼	0.4375 – 28 UNEF	0.21	0.25	0.687
03	⅜	0.562 – 24 UNEF	0.25	0.37	0.812
04	½	0.6878 – 24 UNEF	0.25	0.50	0.937
05	⅝	0.8125 – 20 UNEF	0.31	0.62	1.062
06	¾	0.9375 – 20 UNEF	0.31	0.75	1.187
08	1	1.250 – 18 UNEF	0.37	1.00	1.500
10	1¼	1.5625 – 18 UNEF	0.37	1.25	1.812
12	1½	1.875 – 16 UN	0.43	1.50	2.125
16	2	2.375 – 16 UN	0.43	2.00	2.625
20	2½	2.875 – 16 UN	0.43	2.50	3.125
24	3	3.375 – 16 UN	0.43	3.00	3.625

NOTES:
1. The function of the RP2311 adapter is to attach flexible shielding conduit to rigid shielding conduit (“black iron” pipe).
2. Dimensions are in inches.
3. Manufacturer information provided for historical reference only. Information for procuring material shall be obtained directly from the vendor.

FIGURE 21. RP2311 adapter for SAE AS50151 connector.

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TABLE 21-2. RP2311 manufacture reference.

SAE AS50151 Connector Code			
Conn Code	**Mfr**	**Class**	**Series**
A	ALL MFR		MS-SERIES BACKSHELL
B	AMPHENOL	A	MS3100, MS3101, MS3106, 97-3100, 97-3101, 97-3106
C	AMPHENOL	E, R	MS3100, MS3101, MS3106, 69-3100, 69-3101, 69-3106
D	BENDIX	A, E, R	MS3100, MS3101, MS3106, 10-214, 10-720, 10-721, 10-726
E	CANNON	A	MS3100, MS3101, MS3106, CA3101, CA3106
F	CANNON	E, R	MS3100, MS3101, MS3106

Example:	**M32771-RP2311**	-	**06**	**08**	**01**
	Basic PIN	-	A Thread Code (see table 21-3)	Conduit Size Code (see table 21-1)	Style Code

FIGURE 21. RP2311 adapter for SAE AS50151 connector – Continued.

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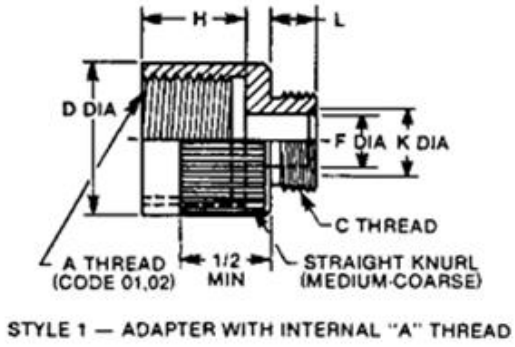
TABLE 21-3. RP2311 connector reference.

A Thread Code	Conn Shell Size	Connector Code (see table 21-2)	A Thread (Class 2B)	H ±0.02	D Dia. ±0.02	E Dia. ±0.02	J (style 01) 1/ ±0.03
01	8S	D	0.375 - 32 UNEF	0.18	0.50	0.28	0.50
02	8S	B	0.4375 - 27 UNS	0.21	0.56	0.34	0.53
03	8	C, E, F	0.4375 - 28 UNEF	0.21	0.56	0.34	0.53
04	8S	A	0.500 - 28 UNEF	0.21	0.62	0.37	0.56
04	10S	A, B, C, D, F	0.500 - 28 UNEF	0.21	0.62	0.37	0.56
05	10S	E	0.5625 - 24 UNEF	0.25	0.68	0.45	0.59
05	10SL	E, F	0.5625 - 24 UNEF	0.25	0.68	0.45	0.59
06	10SL	A, B, C ,D	0.625 - 24 UNEF	0.25	0.75	0.50	0.62
06	12,12S	A, D, F	0.625 - 24 UNEF	0.25	0.75	0.50	0.62
07	12,12S	B, C, E	0.6875 - 24 UNEF	0.25	0.81	0.57	0.65
08	12SL	A	0.750 - 20 UNEF	0.28	0.87	0.62	0.68
08	14,14S	A, B, C, D, E, F	0.750 - 20 UNEF	0.28	0.87	0.62	0.68
09	16,16S	A, B, C, D, E, F	0.875 - 20 UNEF	0.28	1.00	0.75	0.71
10	18	A, B, C, D, E, F	1.000 - 20 UNEF	0.28	1.12	0.87	0.75
11	20	B, D, E, F	1.125 - 18 UNEF	0.28	1.25	1.00	0.78
12	20	C	1.125 - 24 UNS	0.28	1.25	1.00	0.78
13	20,22	A	1.1875 - 18 UNEF	0.28	1.31	1.06	0.81
14	22	B, C, D, E, F	1.250 - 18 UNEF	0.28	1.37	1.12	0.87
15	24	B, C, D, E, F	1.375 - 18 UNEF	0.28	1.50	1.25	1.00
16	24,28	A	1.4375 - 18 UNEF	0.28	1.56	1.31	1.06
17	28	B, C, D, E, F	1.625 - 18 UNEF	0.28	1.75	1.50	1.12
18	32	A	1.750 - 18 UNS	0.28	1.87	1.62	1.18
19	32	D, E, F	1.875 - 16 UN	0.28	2.00	1.75	1.25
20	32	B, C	1.9063 - 18	0.28	2.03	1.78	1.28
21	36	A	2.000 - 18 UNS	0.28	2.12	1.87	1.34
22	36	D	2.0625 - 16 UNS	0.31	2.18	1.93	1.37
23	36	C	2.0625 - 20	0.28	2.18	1.93	1.37
24	36	E, F	2.125 - 16 UN	0.31	2.25	2.00	1.40
25	36	B	2.125 - 18	0.28	2.25	2.00	1.40
26	40	A	2.250 - 16 UN	0.31	2.37	2.12	1.50
27	40	D	2.3125 - 16 UNS	0.31	2.43	2.18	1.56
28	40	B, E, F	2.375 - 16 UN	0.31	2.50	2.25	1.62
29	44	B, F	2.625 - 16 UN	0.31	2.75	2.50	1.87
30	48	F	2.8125 - 18	0.28	2.93	2.68	2.06
31	48	B	2.875 - 16 UN	0.31	3.00	2.75	2.12

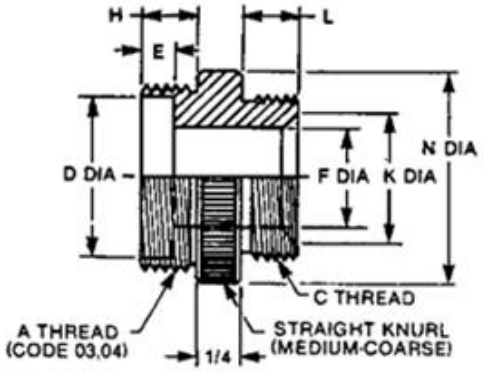
NOTE:
1/ For style 02, J dimension is zero.

FIGURE 21. RP2311 adapter for SAE AS50151 connector – Continued.

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STYLE 1 — ADAPTER WITH INTERNAL "A" THREAD



STYLE 2 — ADAPTER WITH EXTERNAL "A" THREAD

TABLE 22-1. RP2321 adapter sizes.

Standard Conduit		Adapter			
Conduit Size Code	**Conduit Size (Nom. ID)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**	**N Dia. ±0.02**
02	¼	0.4375 - 28 UNEF	0.21	0.25	0.687
03	⅜	0.5625 - 24 UNEF	0.25	0.37	0.812
04	½	0.6875 - 24 UNEF	0.25	0.50	0.937
05	⅝	0.8125 - 20 UNEF	0.31	0.62	1.062
06	¾	0.9375 - 20 UNEF	0.31	0.75	1.187
08	1	1.250 - 18 UNEF	0.37	1.00	1.500
10	1¼	1.5625 - 18 UNEF	0.37	1.25	1.812
12	1½	1.875 - 16 UN	0.43	1.50	2.125
16	2	2.375 - 16 UN	0.43	2.00	2.625
20	2½	2.875 - 16 UN	0.43	2.50	3.125
24	3	3.375 - 16 UN	0.43	3.00	3.625

FIGURE 22. RP2321 adapter for triaxial connector.

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TABLE 22-2. RP2321 manufacture reference.

A Thread Code	Connector	Part No.	A Thread	D Dia.	E	F Dia.	H	Cable (Ref.)	
	ITT/CANNON	Formerly GREMAR						Type	OD
01	152105-2380	5633	0.750 - 16 UNF-2B	0.875	---	0.297	0.562	RG-233/U RG-58C/U	0.216 0.195
02	152100-2390	7991	0.875 - 20 UNEF-2B	1.000	---	0.515	0.500	RG-214/U RG-9B/U	0.425 0.420
03	152190-0000	16312	1.0625 - 18 UNEF-2A	0.922	0.187	0.564	0.312	RG-14A/U RG-293A/U TRF-8 1/	0.545 0.545 0.500
04		11865	1.0625 - 18 UNF-2A	0.922	0.227	0.564	0.335	RG-14A/U RG-293A/U TRF-8 1/	0.545 0.545 0.500

NOTE:
1/ TRF-8 is a specific part number from Times Microwave Systems, Inc., not a part number.

RP2321 PIN Construction:

Example:	**M32771-RP2321**	-	**01**	**08**
	Basic PIN	-	A Thread Code (see table 22-2)	Conduit Size Code (see table 22-1)

NOTES:
1. The function of the RP2321 adapter couples triaxial connectors to RP2100, RP2110, or RP2120 end fittings.
2. RP2321 supersedes RP2320; therefore, any requirement for an RP2320 will be satisfied with an RP2321.
3. Dimensions are in inches.
4. Manufacturer information provided for historical reference only. Information for procuring material shall be obtained directly from the vendor.

FIGURE 22 RP2321 adapter for triaxial connector – Continued.

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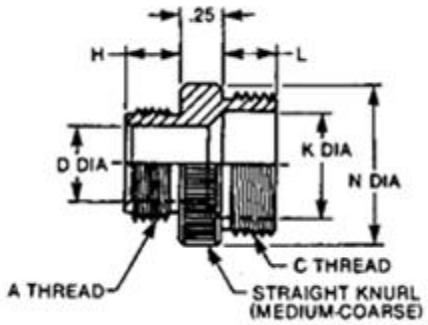


TABLE 23-1. RP2330 conduit reference.

Standard Conduit		Adapter			
Conduit Size Code	**Conduit Size (Nom. ID)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**	**N Dia. ±0.02**
02	¼	0.4375 - 28 UNEF	0.21	0.25	0.687
03	⅜	0.5625 - 24 UNEF	0.25	0.37	0.812
04	½	0.6875 - 24 UNEF	0.25	0.50	0.937
05	⅝	0.8125 - 20 UNEF	0.31	0.62	1.062
06	¾	0.9375 - 20 UNEF	0.31	0.75	1.187
08	1	1.250 - 18 UNEF	0.37	1.00	1.500
10	1¼	1.5625 - 18 UNEF	0.37	1.25	1.812
12	1½	1.875 - 16 UN	0.43	1.50	2.125
16	2	2.375 - 16 UN	0.43	2.00	2.625
20	2½	2.875 - 16 UN	0.43	2.50	3.125
24	3	3.375 - 16 UN	0.43	3.00	3.625

TABLE 23-2. RP2330 coaxial reference.

Adapter (Connector End)				Coaxial Cable (Ref.)	
A Thread Code	**A Thread (Class 2A)**	**H +0.03 -0.00**	**D Dia. +0.02 -0.00**	**Type**	**OD**
02	0.375 - 32 UNEF	0.21	0.226	RG-58, RG-223	0.195, 0.215
04	0.4375 - 28 UNEF	0.25	0.345	RG-5, RG-6, RG-21	0.332
06	0.625 - 24 UNEF	0.31	0.433	RG-9, RG-214	0.420, 0.425
08	0.6875 - 24 UNEF	0.31	0.568	RG-14, RG-294	0.545
10	0.8125 - 20 UNEF	0.31	0.650	RG-57, RG-294	0.625, 0.630
12	1.250 - 18 UNEF	0.31	0.895	RG-17	0.870

FIGURE 23. RP2330 adapter for coaxial connector.

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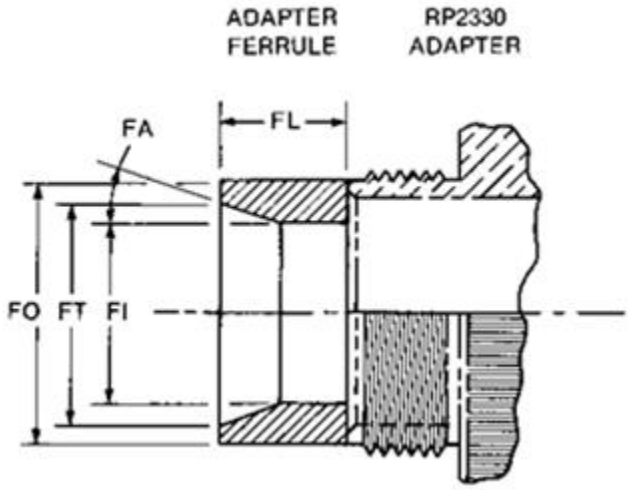


TABLE 23-3. RP2330 ferrule reference.

Adapter Ferrule (for "Wedge-Lock" Design Only)					
A Thread Code (Ref.)	**FO Dia. +0.000 -0.003**	**FI Dia. +0.005 -0.000**	**FT Dia. +0.003 -0.003**	**FA Angle +½° -½°**	**FL Length +0.005 -0.005**
02	---	---	---	---	---
04	0.398	0.235	0.297	24	0.158
06	---	---	---	---	---
08	0.637	0.460	0.530	21.5	0.221
10	0.738	0.568	0.640	18	0.385
12	---	---	---	---	---

RP2330 PIN Construction:

Example:	**M32771-RP2330**	-	**02**	**04**	**N**
	Basic PIN	-	A Thread Code (see table 23-2 and note 3)	Conduit Size Code (see table 23-1)	Material Code

NOTE: No letter in the material code indicates steel. "N" designates nylon (see Note 4).

FIGURE 23. RP2330 adapter for coaxial connector – Continued.

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NOTES:

1. The RP2330 adapter couples coaxial connectors to RP2100, RP2110, or RP2120 end fittings.
2. Select an A thread from table 23-2 that matches that of the back nut on the connector being used. If none of the A threads match, substitute the desired A thread information in lieu of the A thread code in the part number. For example, an adapter with a ½ - 28 thread of 5/16" H-length for a ⅜" conduit would have the PIN M32771-RP2330-1/2-28-5/16-03.
3. If a particular installation requires that the connector body be "floated" off ground, that is, insulated from the conduit, use the material code letter "N" in the PIN. This letter indicates the non-conducting material, nylon.
4. The RP2440 nylon adapter provides an optional method of insulating connector body from conduit.
5. Dimensions are in inches.
6. This RP2330 adapter is designed to replace the gland nut (a simple one-piece threaded component) at the rear of a MIL-PRF-39012 connector, in order to provide a means for terminating flexible shielding conduit. However, MIL-PRF-39012 does not provide configuration control over most of the internal components. As a result, (a) the thread sizes for the rear end of coaxial connectors may vary among manufacturers, and (b) several different internal designs have been produced, some of which are not compatible with the existing RP2330 adapter.

   Because there is not configuration control, the government designation (such as part number M39012/01-0015) for an M39012 connector gives no indication as to which internal design is contained in the connector. For example, the M39012/01-0015, supplied by automatic connector, inc., may have the compatible one-piece gland-nut design or it may have a newer design (called wedge-lock) that includes a gland nut with a captive free-spinning ferrule that presses the fanned-out cable-braid against a conical body without rotating, as the gland nut is tightened. This newer design is not compatible with the existing one-piece RP2330 adapter because the adapter tends to cut the cable-braid in a sawing action as it is tightened.

   As an interim solution to making the RP2330 adapter compatible with the wedge-lock design (for example), an adapter ferrule, to be used in conjunction with the RP2330 adapter, shall be provided. This adapter ferrule should duplicate the basic shape of the free-spinning ferrule contained in the connector.

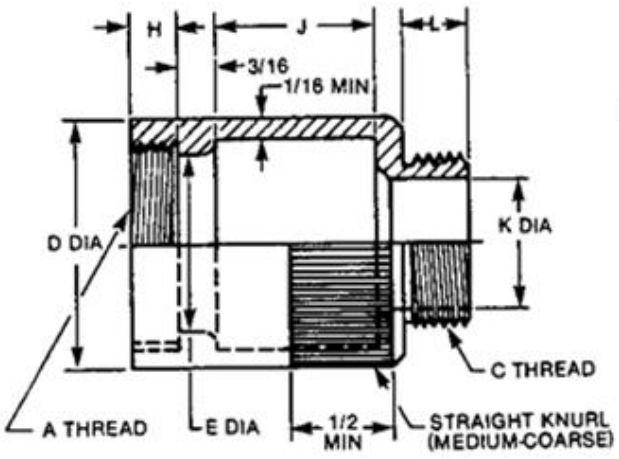
   Table 23-3 lists three sizes of adapter ferrule (for the "wedge-lock" design only) to be used with the RP2330 adapter. Note that the ID of the ferrule restricts the size of coaxial cable diameter that can be used. Note that this ferrule does not provide for retention of the original watertight integrity of the complete connector.

   Another method of obtaining an adapter ferrule is to use the actual ferrule portion of the connector by cutting it from the gland nut.

   A better method of terminating flexible shielding conduit to a coaxial connector may be an adapter designed to attach to the outside of the connector rather than disturbing the internal components of the connector itself. The rear end of the adapter would provide threads for attaching a conduit end fitting.

FIGURE 23. RP2330 adapter for coaxial connector – Continued.

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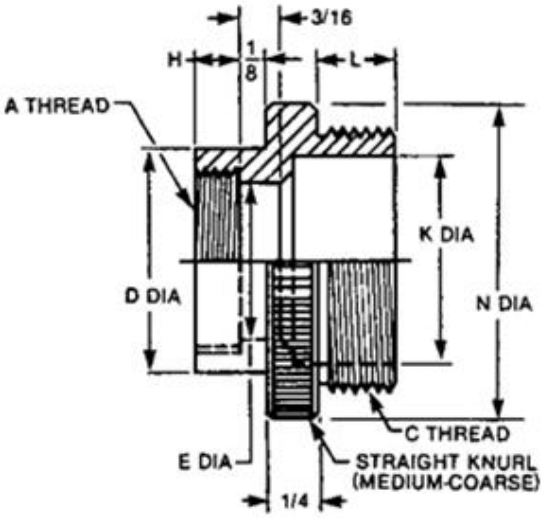


STYLE 01

FOR USE WHEN E DIAMETER IS GREATER THAN K DIAMETER

STYLE 02

SAME AS STYLE 01 EXCEPT J DIMENSION IS ZERO



STYLE 03

FOR USE WHEN E DIAMETER IS EQUAL TO OR LESS THAN K DIAMETER

TABLE 24-1. RP2340 class 2A reference.

Standard Conduit		Adapter			
Conduit Size Code	**Conduit Size (Nom. ID)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**	**N Dia. ±.02**
02	¼	0.4375 - 28 UNEF	0.21	0.25	0.687
03	⅜	0.5625 - 24 UNEF	0.25	0.37	0.812
04	½	0.6875 - 24 UNEF	0.25	0.50	0.937
05	⅝	0.8125 - 20 UNEF	0.31	0.62	1.062
06	¾	0.9375 - 20 UNEF	0.31	0.75	1.187
08	1	1.250 - 18 UNEF	0.37	1.00	1.500
10	1¼	1.5625 - 18 UNEF	0.37	1.25	1.812
12	1½	1.875 - 16 UN	0.43	1.50	2.125
16	2	2.375 - 16 UN	0.43	2.00	2.625
20	2½	2.875 - 16 UN	0.43	2.50	3.125
24	3	3.375 - 16 UN	0.43	3.00	3.625

FIGURE 24. RP2340 adapter for MIL-DTL-26482 connector.

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TABLE 24-2. RP2340 class 2B reference.

A Thread Code	Conn Shell Size	A Thread (Class 2B)	H[2/] ±0.02	D Dia. ±0.02	E Dia. ±0.02	J (Style 01) [1/] ±0.03
07	8	0.4375 - 28 UNEF	[3/]	0.59	0.34	0.87
09	10	0.5625 - 24 UNEF	[3/]	0.71	0.45	0.87
11	12	0.6875 - 24 UNEF	[3/]	0.84	0.57	0.87
13	14	0.8125 - 20 UNEF	0.21	0.96	0.68	0.87
15	16	0.9375 - 20 UNEF	0.21	1.09	0.81	0.87
17	18	1.0625 - 18 UNEF	0.21	1.21	0.93	1.06
19	20	1.1875 - 18 UNEF	[3/]	1.34	1.06	1.18
21	22	1.3125 - 18 UNEF	[3/]	1.46	1.18	1.62
23	24	1.4375 - 18 UNEF	0.21	1.59	1.31	1.68

NOTES:
[1/] For style 02, J dimension is zero.
[2/] The thread length, H, is critical. If a die is used to cut the thread, the H dimension shall be as shown. If a lathe is used to cut the thread, reduce H dimension by 0.06 inch.
[3/] Dimension not available.

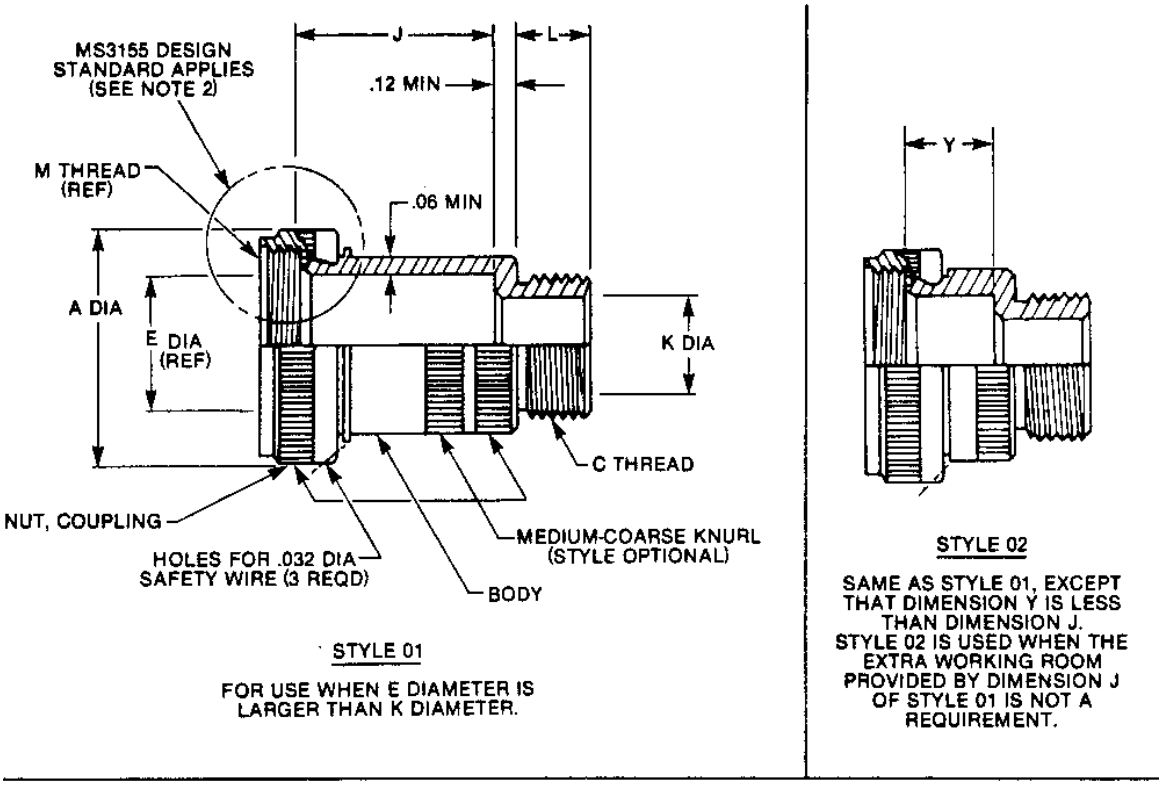
RP2340 PIN Construction:

Example:	**M32771-RP2340**	-	**15**	**08**	**03**
	Basic PIN	-	A Thread Code (see table 24-2)	Conduit Size Code (see table 24-1)	Style Code

NOTES:
1. The RP2340 adapter couples MIL-DTL-26482 connectors (MS3110, MS3111, or MS3116 series) to RP2100, RP2110, or RP2120 end fittings.
2. For MIL-DTL-26482, series 2, the RP2350 adapter may be used.
3. See S9407-AB-HBK-010 for conditions for use of aluminum connector backshells.
4. Dimensions are in inches.

FIGURE 24. RP2340 adapter for MIL-DTL-26482 connector – Continued.

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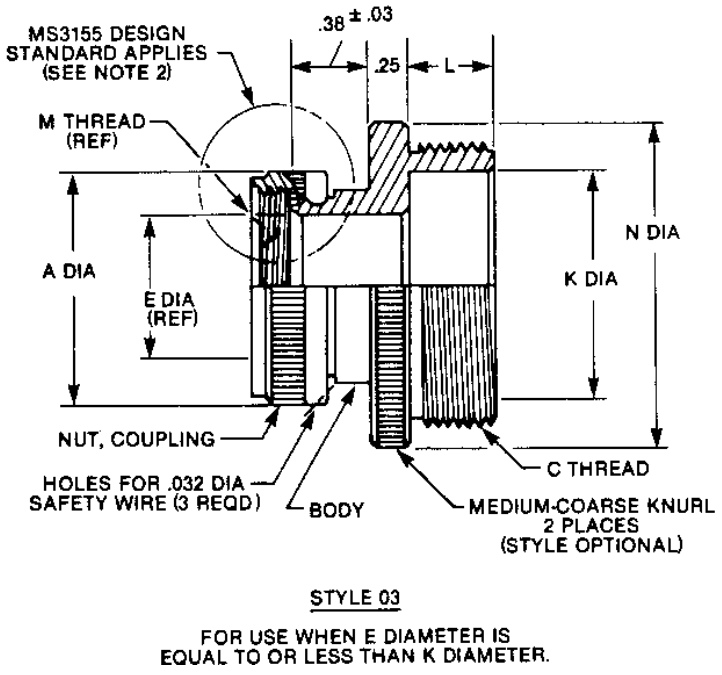


FIGURE 25. RP2350 adapter for SAE AS85049-type connectors.

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TABLE 25-1. RP2350 connector reference.

Conn Dash No.	Connector Shell Size				Adapter				
	MIL-DTL-83723 (Series 1&3)	SAE AS39029 (MS3400 Series) 1/	SAE AS81703 (Series 3)	MIL-DTL-26482 (Series 2) 2/	A Dia. (Max.)	M Thread (Class 2B) (Ref.) Note 3	E Dia. (Ref.) Note 3	J ±0.03 (Style 01)	Y ±0.03 (Style 02)
03	---	---	3	---	0.669	0.562 - 24 UNEF	0.270	1.06	0.62
08	8	8S	---	8	0.617	0.500 - 20 UNF	0.270	1.06	0.62
10	10	10S, 10SL	---	10	0.734	0.625 - 24 UNEF	0.375	1.06	0.69
12	12	12, 12S	7	12	0.858	0.750 - 20 UNEF	0.511	1.06	0.69
14	14	14, 14S	12	14	0.984	0.875 - 20 UNEF	0.585	1.06	0.75
16	16	16, 16S	19	16	1.112	1.000 - 20 UNEF	0.710	1.06	0.75
18	18	18	27	18	1.218	1.062 - 18 UNEF	0.789	1.12	0.75
20	20	20	37	20	1.345	1.188 - 18 UNEF	0.914	1.38	0.81
22	22	22	---	22	1.468	1.312 - 18 UNEF	1.039	1.83	0.81
24	24	24	---	24	1.593	1.438 - 18 UNEF	1.154	1.88	0.81
28	---	28	---	---	1.969	1.750 - 18 UNS	1.389	2.30	0.88
32	---	32	---	---	2.219	2.000 - 18 UNS	1.635	2.55	0.88
36	---	36	---	---	2.469	2.250 - 16 UN	1.850	2.80	0.88
40	---	40	---	---	2.719	2.500 - 16 UN	2.065	3.05	0.94
44	---	44	---	---	2.969	2.750 - 16 UN	2.320	3.55	0.94
48	---	48	---	---	3.219	3.000 - 16 UN	2.570	3.62	0.94
61	---	---	61	---	1.653	1.500 - 18 UNEF	1.194	1.88	0.81

NOTES:

1/ For MS3100, MS3101, or MS3106 connectors, use RP2311 adapter.

2/ For MS3110, MS3111, or MS3116 connectors, use RP2340 adapter.

FIGURE 25. RP2350 adapter for SAE AS85049-type connectors – Continued.

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TABLE 25-2. RP2350 conduit reference.

Standard Conduit		Adapter			
Conduit Size Code	**Conduit Size (Nom. ID)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**	**N Dia. ±0.02**
02	¼	0.4375 - 28 UNEF	0.21	0.25	0.687
03	⅜	0.5625 - 24 UNEF	0.25	0.37	0.812
04	½	0.6875 - 24 UNEF	0.25	0.50	0.937
05	⅝	0.8125 - 20 UNEF	0.31	0.62	1.062
06	¾	0.9375 - 20 UNEF	0.31	0.75	1.187
08	1	1.250 - 18 UNEF	0.37	1.00	1.500
10	1¼	1.5625 - 18 UNEF	0.37	1.25	1.812
12	1½	1.875 - 16 UN	0.43	1.50	2.125
16	2	2.375 - 16 UN	0.43	2.00	2.625
20	2½	2.875 - 16 UN	0.43	2.50	3.125
24	3	3.375 - 16 UN	0.43	3.00	3.625

RP2350 PIN Construction:

Example:	**M32771-RP2350**	-	**15**	**06**	**01**	**A**
	Basic PIN	-	Connector Dash No. (see table 25-1)	Conduit Size Code (see table 25-2)	Style Code	Material Code

NOTE: No letter in the material code indicates steel. "A" designates aluminum alloy (see note 5).

NOTES:

1. The RP2350 adapter couples connectors listed in table 25-1 to RP2100, RP2110, or RP2120 conduit end fittings.
2. The coupling nut end of this adapter shall be as specified in SAE AS85049. SAE AS85049 takes precedence over this specification.
3. The E diameter and M thread of this specification are the same as the E diameter and M thread of SAE AS85049 and are included for reference purposes.
4. Aluminum alloy may be specified provided that minimum spacing requirements for the adapter (backshell), delineated in S9407-AB-HBK-010, are met.
5. Dimensions are in inches.

FIGURE 25. RP2350 adapter for SAE AS85049-type connectors – Continued.

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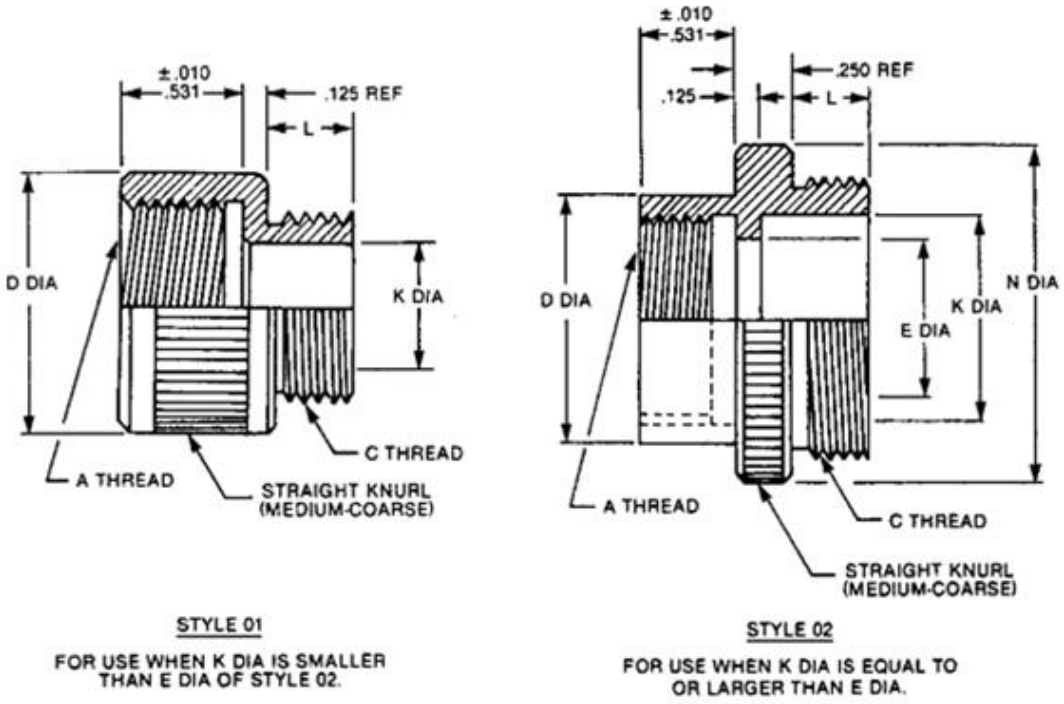


TABLE 26-1. RP2360 class 2A reference.

Standard Conduit		Adapter			
Conduit Size Code	**Conduit Size (Nom. ID)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**	**N Dia. ±0.02**
02	¼	0.4375 - 28 UNEF	0.21	0.25	0.687
03	⅜	0.5625 - 24 UNEF	0.25	0.37	0.812
04	½	0.6875 - 24 UNEF	0.25	0.50	0.937
05	⅝	0.8125 - 20 UNEF	0.31	0.62	1.062
06	¾	0.9375 - 20 UNEF	0.31	0.75	1.187
08	1	1.250 - 18 UNEF	0.37	1.00	1.500
10	1¼	1.5625 - 18 UNEF	0.37	1.25	1.812
12	1½	1.875 - 16 UN	0.43	1.50	2.125
16	2	2.375 - 16 UN	0.43	2.00	2.625
20	2½	2.875 - 16 UN	0.43	2.50	3.125
24	3	3.375 - 16 UN	0.43	3.00	3.625

FIGURE 26. RP2360 adapter for MIL-DTL-28840 backshell.

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TABLE 26-2. RP2360 class 2B reference.

A Thread Code	Shell Size (Ref.)	A Thread (Class 2B)	D Dia. ±0.015	E Dia. ±0.015
01	B, C	0.625-24 UNEF	0.875	0.453
02	B, C	0.750-20 UNEF	1.000	0.578
03	C	0.875-20 UNEF	1.125	0.703
04	D, E	1.000-20 UNEF	1.250	0.828
05	C, D, F	1.118-18 UNEF	1.438	1.000
06	D, E, F, G	1.438-18 UNEF	1.688	1.250
07	F, G, H	1.750-18 UNS	2.000	1.562
08	G, H, J	2.000-18 UNS	2.250	1.813
09	J	2.250-16 UNS	2.500	2.062
10	C, D, E	1.125-18 UNEF	1.375	.938
11	C, D, E, F, G	1.250-18 UNEF	1.500	1.078
12	D, E, F, G	1.625-18 UNEF	1.875	1.438
13	F, G, H	1.875-18 UNS	2.125	1.688
14	G, H, J	2.125-16 UN	2.375	1.938
15	J	2.750-16 UN	3.000	2.563

RP2360 PIN Construction:

Example:	**M32771-RP2360**	-	**06**	**08**	**01**
	Basic PIN	-	A Thread Code (see table 26-2)	Conduit Size Code (see table 26-1)	Style Code

NOTES:

1. The RP2360 adapter provides a non-environmental coupling between the MIL-DTL-28840 backshells and the RP2100, RP2110, or RP2120 end fittings.
   Specifically, the A thread of this adapter (A thread codes 01 through 09) mates with the V thread of the M28840/6 backshell (straight), the M23340/8 backshell (90°), or the M28840/9 backshell (45°).
   The A thread codes 10 through 15 mate with the C threads of M28840/5, M28840/25, or M28840/27. (The A thread codes 03 and 04 mate with these C threads also.)
2. This adapter will also mate with the plug assemblies M28840/26, M28840/28, or M28840/29 since the backshells used in these plug assemblies are the M28840/6, M28840/8, and M28840/9 types, respectively.
3. The column labeled “Shell Size (Ref.)” is presented for reference purposes only. It does not directly affect the selection of an adapter size to mate with a particular backshell size. Only the C thread or the V thread size of a backshell determines which A thread size of adapter to use.
4. See S9407-AB-HBK-010 for conditions for use of aluminum connector backshells.
5. Dimensions are in inches.

FIGURE 26. RP2360 adapter for MIL-DTL-28840 backshell – Continued.

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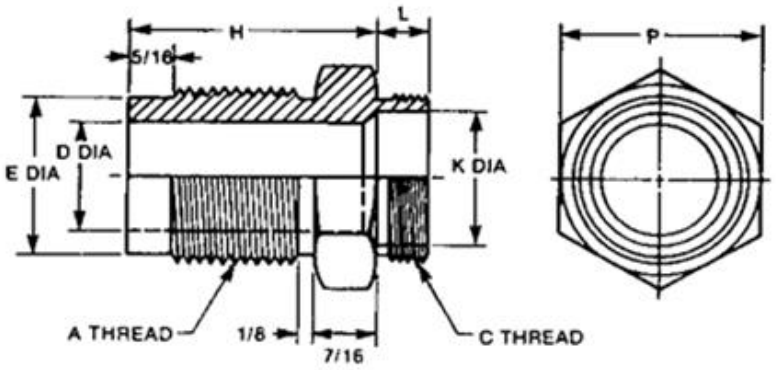


TABLE 27-1. RP2400 conduit reference.

Standard Conduit		Adapter		
Conduit Size Code	**Conduit Size (Nom. ID)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**
02	¼	0.4375 - 28 UNEF	0.21	0.25
03	⅜	0.5625 - 24 UNEF	0.25	0.37
04	½	0.6875 - 24 UNEF	0.25	0.50
05	⅝	0.8125 - 20 UNEF	0.31	0.62
06	¾	0.9375 - 20 UNEF	0.31	0.75
08	1	1.250 - 18 UNEF	0.37	1.00
10	1¼	1.5625 - 18 UNEF	0.37	1.25
12	1½	1.875 - 16 UN	0.43	1.50
16	2	2.375 - 16 UN	0.43	2.00
20	2½	2.875 - 16 UN	0.43	2.50
24	3	3.375 - 16 UN	0.43	3.00

TABLE 27-2. RP2400 stuffing tube reference.

Adapter						MIL-S-24235/1
A Thread Code	**A Thread (Class 2A)**	**D Dia. ±0.005**	**E Dia. ±0.005**	**H Dim. ±0.02**	**P Dim. ±0.02**	**Stuffing Tube Size**
02	1.125 - 12 UNF	0.609	0.984	1.56	1.25	1
05	1.3125 - 12 UN	0.812	1.171	1.56	1.43	2
08	1.625 - 12 UN	1.062	1.484	1.56	1.78	3
11	1.8125 - 12 UN	1.265	1.671	1.87	1.93	4
12	2.0625 - 12 UN	1.437	1.921	2.00	2.25	5
14	2.3125 - 12 UN	1.625	2.173	2.12	2.46	6
15	2.6875 - 12 UN	1.875	2.531	2.31	2.81	7
17	3.000 - 12 UN	2.187	2.850	2.31	3.09	8
18	3.125 - 12 UN	2.312	2.968	2.37	3.28	9

FIGURE 27. RP2400 adapter for MIL-S-24235/1 stuffing tube.

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RP2400 PIN Construction:

Example:	**M32771-RP2400**	-	**12**	**10**
	Basic PIN	-	A Thread Code (see table 27-2)	Conduit Size Code (see table 27-1)

NOTES:
1. The RP2400 adapter couples MIL-S-24235/1 stuffing tubes to RP2100, RP2110, or RP2120 end fittings.
2. The maximum "P" dimension of table 27-2 should be larger than the "C" thread OD when the required "C" and "A" thread combination would result in "C" OD larger than the "P" dimension given for the required "A" thread OD.
3. Dimensions are in inches.

FIGURE 27 RP2400 adapter for MIL-S-24235/1 stuffing tube – Continued.

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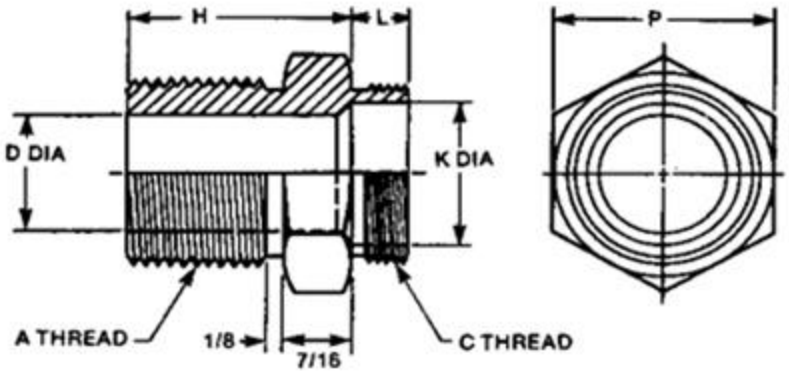


TABLE 28-1. RP2411 C thread reference.

Standard Conduit		Adapter		
Conduit Size Code	Conduit Size (Nom. ID)	C Thread (Class 2A)	L ±0.02	K Dia. (Min.)
02	¼	0.4375 – 28 UNEF	0.21	0.25
03	⅜	0.5625 – 24 UNEF	0.25	0.37
04	½	0.6875 – 24 UNEF	0.25	0.50
05	⅝	0.8125 – 20 UNEF	0.31	0.62
06	¾	0.9375 – 20 UNEF	0.31	0.75
08	1	1.250 – 18 UNEF	0.37	1.00
10	1¼	1.5625 – 18 UNEF	0.37	1.25
12	1½	1.875 – 16 UN	0.43	1.50
16	2	2.375 – 16 UN	0.43	2.00
20	2½	2.875 – 16 UN	0.43	2.50
24	3	3.375 – 16 UN	0.43	3.00

Table 28-2. RP2411 A thread reference.

Adapter				
A Thread Code	A Thread (Class 2A)	D Dia. ±0.005	H Dim. ±0.02	P Dim. ±0.02
01	1⅛ - 14 N	0.500	1.25	1.25
03	1.288 – 14 N	0.750	1.25	1.43
04	¾ - 14 NPSM	0.750	1.25	1.37
06	1 - 11½ NPSM	1.000	1.25	1.68
07	1¼ - 11½ NPSM	1.000	1.25	1.81
09	1¼ - 11½ NPSM	1.180	1.25	1.81
10	2.025 - 11½ N	1.250	1.56	2.18
13	2¼ - 11½ N	1.500	1.69	2.50
16	2½ - 8 NPSM	2.000	2.00	3.00
19	3 - 8 NPSM	2.500	2.06	3.62
20	1½ - 11½ NPSM	1.500	1.25	2.50
21	2 - 11½ NPSM	2.000	1.25	3.00

RP2411 PIN Construction:

Example:	**M32771-RP2411**	-	**04**	**08**
	Basic PIN	-	A Thread Code (see table 28-2)	Conduit Size Code (see table 28-1)

NOTES:
1. The RP2411 adapter couples miscellaneous fittings to RP2100, RP2110, or RP2120 end fittings.
2. The American National thread series (N) has been superseded by the Unified thread series (UN). The series N thread sizes listed in table 28-2 are retained for use with those items which have series N threads.
3. Dimensions are in inches.

FIGURE 28. RP2411 adapter for miscellaneous threaded fittings.

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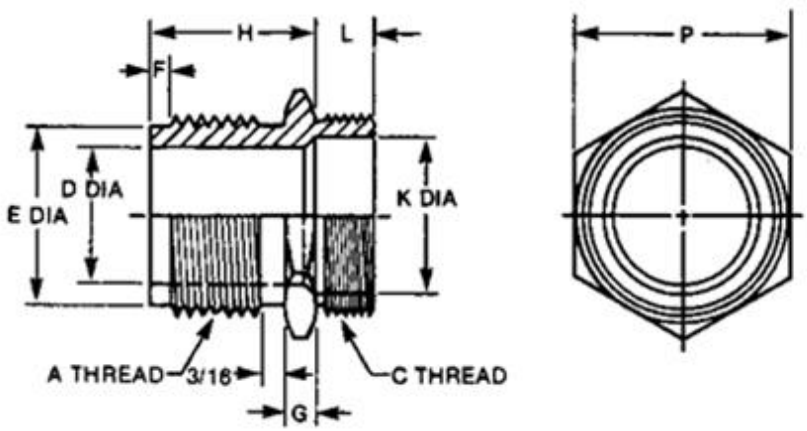


TABLE 29-1. RP2420 conduit reference.

Standard Conduit		Adapter		
Conduit Size Code	**Conduit Size (Nom. ID)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**
02	¼	0.4375 – 28 UNEF	0.21	0.25
03	⅜	0.5625 – 24 UNEF	0.25	0.37
04	½	0.6875 – 24 UNEF	0.25	0.50
05	⅝	0.8125 – 20 UNEF	0.31	0.62
06	¾	0.9375 – 20 UNEF	0.31	0.75
08	1	1.250 – 18 UNEF	0.37	1.00
10	1¼	1.5625 – 18 UNEF	0.37	1.25
12	1½	1.875 – 16 UN	0.43	1.50
16	2	2.375 – 16 UN	0.43	1.50
20	2½	2.875 – 16 UN	0.43	2.50
24	3	3.375 – 16 UN	0.43	3.00

FIGURE 29. RP2420 adapter for MIL-DTL-24231 hull fitting.

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TABLE 29-2. RP2420 hull fitting reference.

Adapter								Hull Fitting Assembly MIL-DTL-24231
A Thread Code	A Thread (Class 2A)	D Dia. ±0.005	E Dim. ±0.02	F Dim. ±0.02	G Dim. ±0.02	H Dim. ±0.02	P Dim. ±0.02	
01	1.250 – 12 UNF	0.875	1.13	0.12	0.18	1.06	1.37	5, 7
02	1.000 – 12 UNF	0.515	---	0.00	0.18	0.75	1.12	6, 8
03	1.250 – 12 UNF	0.671	---	0.00	0.18	0.87	1.37	9, 24
04	1.250 – 12 UNF	0.796	---	0.00	0.18	0.87	1.37	9
05	1.750 – 12 UN	1.312	1.63	0.12	0.18	1.06	1.87	10, 11
06	2.652 – 12 UN	2.062	---	0.00	0.25	1.81	2.81	19

RP2420 PIN Construction:

Example:	**M32771-RP2420**	-	**05**	**12**
	Basic PIN	-	A Thread Code (see table 29-2)	Conduit Size Code (see table 29-1)

NOTES:
1. The RP2420 adapter couples MIL-DTL-24231 pressureproof connectors (hull fittings) to RP2100, RP2110, or RP2120 end fittings.
2. Dimensions are in inches.

FIGURE 29 RP2420 adapter for MIL-DTL-24231 hull fitting – Continued.

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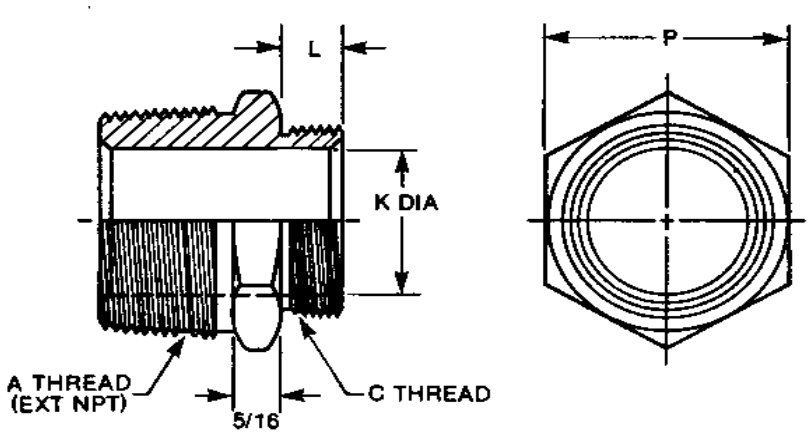


TABLE 30-1. RP2430 reference.

Standard Conduit		Adapter				
Conduit Size Code	**Conduit Size (Nom. ID)**	**A Pipe Thread (Class 2A)**	**P (Min.)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**
02	¼	¼ - 18 NPT	0.69	0.4375 – 28 UNEF	0.21	0.25
03	⅜	⅜ - 18 NPT	0.81	0.5625 – 24 UNEF	0.25	0.37
04	½	½ - 14 NPT	1.00	0.6875 – 24 UNEF	0.25	0.50
05	⅝	¾ - 14 NPT	1.18	0.8125 – 20 UNEF	0.31	0.62
06	¾	¾ - 14 NPT	1.18	0.9375 – 20 UNEF	0.31	0.75
08	1	1 - 11½ NPT	1.50	1.250 – 18 UNEF	0.37	1.00
10	1¼	1¼ - 11½ NPT	1.81	1.5625 – 18 UNEF	0.37	1.25
12	1½	1½ - 11½ NPT	2.12	1.875 – 16 UN	0.43	1.50
16	2	2 - 11½ NPT	2.62	2.375 – 16 UN	0.43	2.00
20	2½	2½ - 8 NPT	3.00	2.875 – 16 UN	0.43	2.50
24	3	3 - 8 NPT	3.62	3.375 – 16 UN	0.43	3.00

RP2430 PIN Construction:

Example: **M32771-RP2430** - **06**

Basic PIN - Conduit Size Code (see table 30-1)

NOTES:
1. The RP2430 adapter couples any fitting having a female tapered pipe thread to RP2100, RP2110, or RP2120 end fittings.
2. Dimensions are in inches.

FIGURE 30. RP2430 adapter for pipe thread (tapered).

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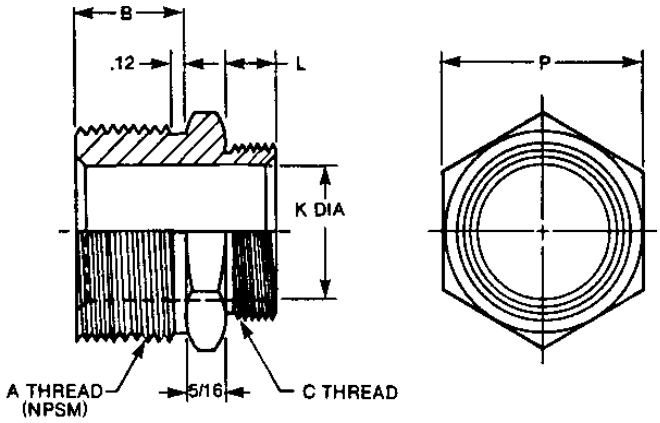


TABLE 31-1. RP2431 reference.

Standard Conduit		Adapter					
Conduit Size Code	**Conduit Size (Nom. ID)**	**A Pipe Thread (Class 2A)**	**B Dim. ±0.02**	**P (Min.)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**
02	¼	¼ - 18 NPSM	0.75	0.69	0.4375 – 28 UNEF	0.21	0.25
03	⅜	⅜ - 18 NPSM	0.75	0.81	0.5625 – 24 UNEF	0.25	0.37
04	½	½ - 14 NPSM	0.87	1.00	0.6875 – 24 UNEF	0.25	0.50
05	⅝	¾ - 14 NPSM	0.87	1.18	0.8125 – 20 UNEF	0.31	0.62
06	¾	¾ - 14 NPSM	0.87	1.18	0.9375 – 20 UNEF	0.31	0.75
08	1	1 - 11½ NPSM	0.87	1.50	1.250 – 18 UNEF	0.37	1.00
10	1¼	1¼ - 11½ NPSM	0.87	1.81	1.5625 – 18 UNEF	0.37	1.25
12	1½	1½ - 11½ NPSM	1.00	2.12	1.875 – 16 UN	0.43	1.50
16	2	2 - 11½ NPSM	1.19	2.62	2.375 – 16 UN	0.43	2.00
20	2½	2½ - 8 NPSM	1.62	3.00	2.875 – 16 UN	0.43	2.50
24	3	3 - 8 NPSM	1.69	3.62	3.375 – 16 UN	0.43	3.00

RP2431 PIN Construction:

Example: **M32771-RP2431** - **06**

Basic PIN - Conduit Size Code (see table 31-1)

NOTES:
1. The RP2431 adapter couples any fitting having a female straight pipe thread to RP2100, RP2110, or RP2120 end fittings.
2. Dimensions are in inches.

FIGURE 31. RP2431 adapter for pipe thread (straight).

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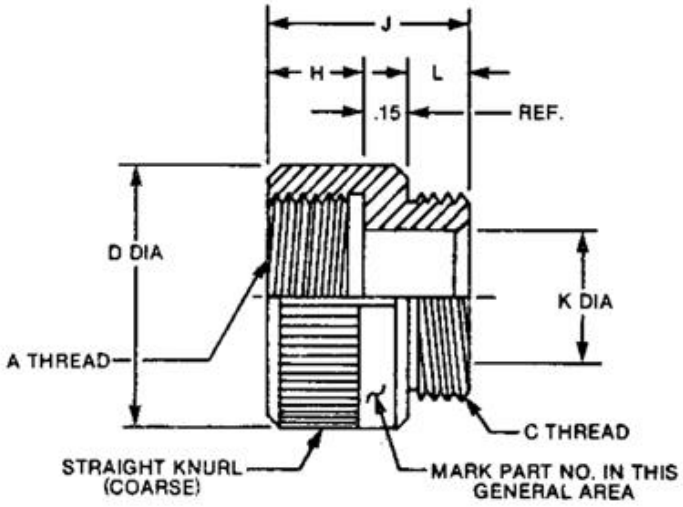


TABLE 32-1. RP2440 sizing.

Standard Conduit		Adapter						
Conduit Size Code	**Conduit Size (Nom. ID)**	**D Dia. ±0.02**	**H +0.02 -0.00**	**A Thread (Class 2B)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**	**J ±0.03**
02	¼	0.75	0.24	0.4375 - 28 UNEF	0.4375 - 28 UNEF	0.21	0.25	0.60
03	⅜	0.88	0.28	0.5625 - 24 UNEF	0.5625 - 24 UNEF	0.25	0.37	0.68
04	½	1.00	0.28	0.6875 - 24 UNEF	0.6875 - 24 UNEF	0.25	0.50	0.68
05	⅝	1.12	0.34	0.8125 - 20 UNEF	0.8125 - 20 UNEF	0.31	0.62	0.80
06	¾	1.25	0.34	0.9375 - 20 UNEF	0.9375 - 20 UNEF	0.31	0.75	0.80
08	1	1.56	0.40	1.250 - 18 UNEF	1.250 - 18 UNEF	0.37	1.00	0.92
10	1¼	1.88	0.40	1.5625 - 18 UNEF	1.5625 - 18 UNEF	0.37	1.25	0.92
12	1½	2.19	0.46	1.875 - 16 UN	1.875 - 16 UN	0.43	1.50	1.04
16	2	2.69	0.46	2.375 - 16 UN	2.375 - 16 UN	0.43	2.00	1.04
20	2½	3.19	0.46	2.875 - 16 UN	2.875 - 16 UN	0.43	2.50	1.04
24	3	3.69	0.46	3.375 - 16 UN	3.375 - 16 UN	0.43	3.00	1.04

FIGURE 32. RP2440 adapter, nonmetallic.

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TABLE 32-2. RP2440 material reference.

Material Code	Material	Specification
N	Nylon, rigid: rods, tubes, etc.	L-P-410
	Nylon, molding plastic	ASTM D4066 Type II
STN	Super tough nylon ZYTEL STB01 (or equal)	Dupont spec. sheets or equivalent source

RP2440 PIN Construction:

Example:	**M32771-RP2440**	-	**06**	**N**
	Basic PIN	-	Conduit Size Code (see table 32-1)	Material Code (see table 32-2)

NOTES:
1. The RP2440 adapter couples an RP2000-series metallic end fitting to an RP2000-series metallic adapter and provides electrical isolation between them.
2. Dimensions are in inches.

FIGURE 32 RP2440 adapter, nonmetallic – Continued.

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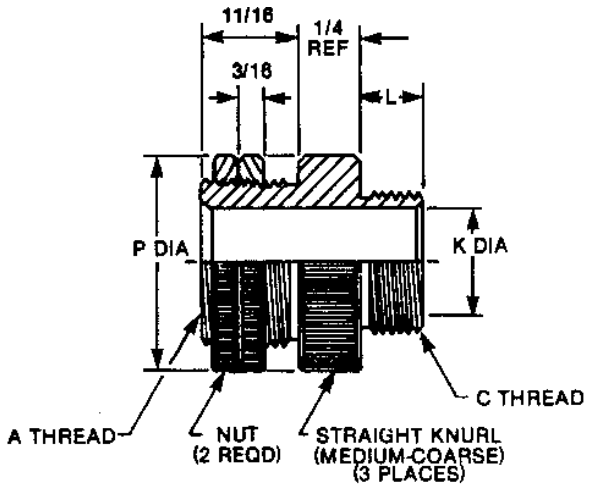


TABLE 33-1. RP2500 sizing.

Standard Conduit		Adapter				
Conduit Size Code	**Conduit Size (Nom. ID)**	**A Thread (Class 2A)**	**P Dia. (Min.)**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**
02	¼	0.6250 - 24 UNEF	0.87	0.4375 - 28 UNEF	0.21	0.25
03	⅜	0.750 - 20 UNEF	1.00	0.5625 - 24 UNEF	0.25	0.37
04	½	0.875 - 20 UNEF	1.12	0.6875 - 24 UNEF	0.25	0.50
05	⅝	1.000 - 20 UNEF	1.25	0.8125 - 20 UNEF	0.31	0.62
06	¾	1.1875 - 18 UNEF	1.43	0.9375 - 20 UNEF	0.31	0.75
08	1	1.4375 - 18 UNEF	1.68	1.250 - 18 UNEF	0.37	1.00
10	1¼	1.750 - 16 UN	2.00	1.5625 - 18 UNEF	0.37	1.25
12	1½	2.000 - 16 UN	2.25	1.875 - 16 UN	0.43	1.50
16	2	2.500 - 16 UN	3.00	2.375 - 16 UN	0.43	2.00
20	2½	3.000 - 16 UN	3.50	2.875 - 16 UN	0.43	2.50
24	3	3.500 - 16 UN	4.00	3.375 - 16 UN	0.43	3.00

RP2500 PIN Construction:

Example: **M32771-RP2500** - **06**

Basic PIN - Conduit Size Code (see table 33-1)

NOTES:
1. The RP2500 adapter provides the capability of terminating an RP2100, RP2110, or RP2120 end fitting to a panel.
2. Dimensions are in inches.

FIGURE 33. RP2500 adapter for panel termination.

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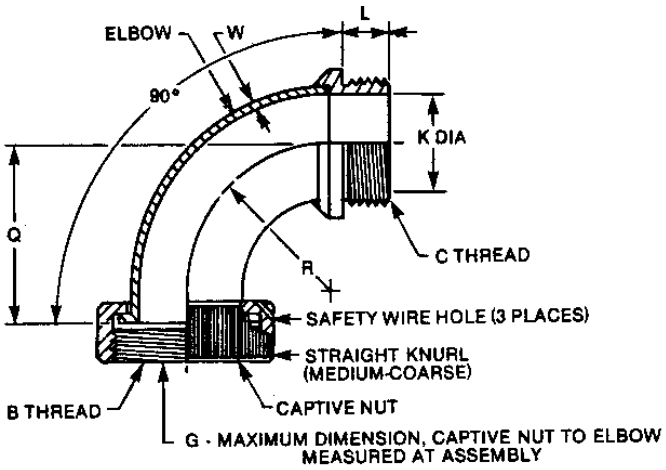


TABLE 34-1. RP2610 sizing.

Standard Conduit		Adapter							
Conduit Size Code	**Conduit Size (Nom. ID)**	**B Thread (Class 2B)**	**G +0.01 -0.02**	**Q ±0.06**	**R Bend Radius ±0.06**	**W Min. Wall Thickness**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**
02	¼	0.4375 - 28 UNEF	0.14	0.81	0.43	0.028	0.4375 - 28 UNEF	0.21	0.25
03	⅜	0.5625 - 24 UNEF	0.17	0.93	0.56	0.028	0.5625 - 24 UNEF	0.25	0.37
04	½	0.6875 - 24 UNEF	0.17	1.12	0.75	0.028	0.6875 - 24 UNEF	0.25	0.50
05	⅝	0.8125 - 20 UNEF	0.23	1.25	0.87	0.028	0.8125 - 20 UNEF	0.31	0.62
06	¾	0.9375 - 20 UNEF	0.23	1.47	1.06	0.028	0.9375 - 20 UNEF	0.31	0.75
08	1	1.250 - 18 UNEF	0.29	1.81	1.37	0.028	1.250 - 18 UNEF	0.37	1.00
10	1¼	1.5625 - 18 UNEF	0.29	2.12	1.68	0.028	1.5625 - 18 UNEF	0.37	1.25
12	1½	1.875 - 16 UN	0.35	2.50	2.06	0.058	1.875 - 16 UN	0.43	1.50
16	2	2.375 - 16 UN	0.35	3.12	2.68	0.058	2.375 - 16 UN	0.43	2.00
20	2½	2.875 - 16 UN	0.35	3.75	3.31	0.058	2.875 - 16 UN	0.43	2.50
24	3	3.375 - 16 UN	0.35	5.00	4.50	0.058	3.375 - 16 UN	0.43	3.00

RP2610 PIN Construction:

Example: **M32771-RP2610** - **06**

Basic PIN - Conduit Size Code (see table 34-1)

NOTES:
1. The RP2610 adapter converts a straight RP2100 end fitting into a 90° end fitting. See RP2110.
2. Dimensions are in inches.

FIGURE 34. RP2610 adapter, 90°.

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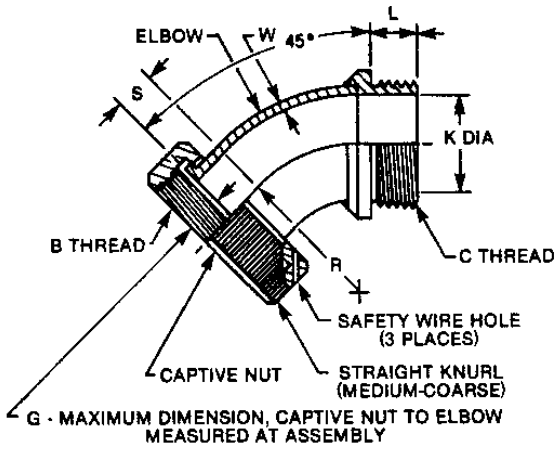


TABLE 35-1. RP2620 sizing.

Standard Conduit		Adapter							
Conduit Size Code	**Conduit Size (Nom. ID)**	**B Thread (Class 2B)**	**G +0.01 -0.02**	**R Bend Radius ±0.06**	**S ±0.06**	**W Min. Wall Thickness**	**C Thread (Class 2A)**	**L ±0.02**	**K Dia. (Min.)**
02	¼	0.4375 - 28 UNEF	0.14	0.43	0.37	0.028	0.4375 - 28 UNEF	0.21	0.25
03	⅜	0.5625 - 24 UNEF	0.17	0.56	0.37	0.028	0.5625 - 24 UNEF	0.25	0.37
04	½	0.6875 - 24 UNEF	0.17	0.75	0.37	0.028	0.6875 - 24 UNEF	0.25	0.50
05	⅝	0.8125 - 20 UNEF	0.23	0.87	0.37	0.028	0.8125 - 20 UNEF	0.31	0.62
06	¾	0.9375 - 20 UNEF	0.23	1.06	0.37	0.028	0.9375 - 20 UNEF	0.31	0.75
08	1	1.250 - 18 UNEF	0.29	1.37	0.43	0.028	1.250 - 18 UNEF	0.37	1.00
10	1¼	1.5625 - 18 UNEF	0.29	1.68	0.43	0.028	1.5625 - 18 UNEF	0.37	1.25
12	1½	1.875 - 16 UN	0.35	2.06	0.43	0.058	1.875 - 16 UN	0.43	1.50
16	2	2.375 - 16 UN	0.35	2.68	0.43	0.058	2.375 - 16 UN	0.43	2.00
20	2½	2.875 - 16 UN	0.35	3.31	0.43	0.058	2.875 - 16 UN	0.43	2.50
24	3	3.375 - 16 UN	0.35	4.50	0.50	0.058	3.375 - 16 UN	0.43	3.00

RP2620 PIN Construction:

Example: **M32771-RP2620** - **06**

Basic PIN - Conduit Size Code (see table 35-1)

NOTES:
1. The RP2620 adapter converts a straight RP2100 end fitting into a 45° end fitting. See RP2120.
2. Dimensions are in inches.

FIGURE 35. RP2620 adapter, 45°.

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CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
Air Force – 99
DLA – GS

Review activities:
Army – CR, MI
Navy – AS, CG, YD
Air Force – 03, 11, 19, 84, 85
DLA – GS6

Civil agencies:
GSA – FAS

Preparing activity:
Navy – SH
(Project 5975-2023-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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